

Profile-Driven Simulation and Hardware-Aware Training for Organic Optoelectronic Compute-in-Memory Vision

by

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Abstract

Compute-in-memory (CIM) architectures reduce the data-movement bottleneck of conventional digital systems by performing matrix–vector multiplications directly inside memory arrays. Among emerging device routes, organic optoelectronic devices offer a distinctive physical substrate for neuromorphic vision systems through multi-level conductance tunability, low-voltage operation, and intrinsic photosensitivity. Translating these device-level advantages into system-level visual performance, however, remains constrained by device variability, limited analog-to-digital converter (ADC) resolution, retention drift, and transfer across fresh hardware instances.

This thesis develops a reusable hardware-aware training (HAT) simulation framework for organic optoelectronic CIM arrays supporting vision transformers and convolutional neural networks under realistic device non-idealities. The framework maps literature-reported device characteristics, including conductance window, cycle-to-cycle variability, device-to-device mismatch, retention dynamics, photoresponse nonlinearity, and write nonlinearity, into task-level visual accuracy.

The experiments first evaluate ResNet-18, ConvNeXt-Tiny, and Tiny-ViT-5M across CIFAR-10, CIFAR-100, Flowers-102, and ImageNet-1k baselines. Building on these baselines, the thesis introduces a taxonomy of HAT regimes and identifies hardware-instance overfitting: standard fixed-mask HAT can perform well on the training-time hardware instance yet collapse to near-chance accuracy on fresh mismatch realizations. Ensemble HAT addresses this failure by resampling structured device-to-device mismatch fields at epoch boundaries, recovering CIFAR-10 fresh-instance accuracy to $86.16 \pm 0.19\%$ across three training seeds.

The thesis further constructs a failure-mode atlas for organic optoelectronic CIM deployment, covering hardware-instance overfitting, severe write-nonlinearity collapse, the limited diagnostic ceiling of MLP-only linearization, bounded degradation under spatially correlated mismatch, and retention-drift plateaus. These results provide system-level decision support for algorithm–device co-design and clarify which training interventions transfer to new hardware instances and which remain constrained by physical modeling assumptions.

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Chapter 1

Hardware-Instance Overfitting and the Ensemble HAT Principle

1.1 Introduction: the deployment gap in analog CIM

Organic optoelectronic compute-in-memory (CIM) systems are invariably introduced through device-level figures of merit: conductance window, retention time, write asymmetry, optical responsivity. Those metrics are necessary, but they are not sufficient to answer the system question that matters for deployment: *what accuracy survives once a modern vision backbone is mapped to an analog array with structured mismatch and finite readout precision?* This thesis treats that question as a materials-to-system translation problem rather than a pure device problem.

The starting point is the canonical hardware-aware training (HAT) regime. Under nominal evaluation on the hardware instance seen during training, HAT appears effective: the hybrid Tiny-ViT recovers most of the source-domain accuracy lost under the canonical analog noise model. However, that apparent success is misleading. When the trained model is evaluated on a fresh hardware instance—a new fixed device-to-device (D2D) mismatch realization drawn from the same distribution—the standard HAT model collapses to chance level. This exposes the central theme of the thesis: robustness to analog noise is not the same thing as robustness to hardware-instance shift.

The contributions of this thesis are threefold. First, a reusable simulation framework for hardware-aware training of vision transformers on analog CIM arrays with realistic device non-idealities. Second, a taxonomy of HAT regimes and the discovery of *hardware-instance overfitting*—the phenomenon that fixed-mask training produces models which memorise a specific device mismatch realisation and collapse to chance level on fresh hardware instances. Third, a deployment envelope that systematically maps which combinations of noise models and training interventions transfer to new

hardware instances and which do not, including rigorous negative results that falsify natural mitigation hypotheses.

This chapter focuses on two linked questions. Why does fixed-mask HAT fail so catastrophically on fresh arrays despite performing well on the training-time array? And what training intervention restores transfer without changing the deployment hardware assumptions? The answer is Ensemble HAT: resampling the structured D2D mismatch field at epoch boundaries during training so that the model learns over a distribution of arrays rather than a single memorised instance.

1.2 The empirical signature: from source-domain rescue to fresh-instance collapse

Analog-deployment experiments must distinguish between three regimes: source-domain accuracy on the training-time hardware instance, repeated Monte Carlo accuracy on the *same* instance, and fresh-instance transfer on newly sampled hardware realisations. Only the third quantity answers the practical deployment question. The distinction is not academic: under the canonical uniform-noise protocol ($\sigma_{\text{D2D}} = 10\%$, $\sigma_{\text{C2C}} = 5\%$, $NL = 1.0$), fixed-mask HAT reaches $87.95 \pm 0.27\%$ on the training-time instance but collapses to $10.00 \pm 0.00\%$ on fresh instances. The 77.95 pp gap is the empirical cost of hardware-instance overfitting.

The decisive observation is simple. A standard HAT model trained with a single fixed D2D mask reaches high source-domain accuracy, yet its fresh-instance accuracy collapses to approximately 10%—the single-class baseline for CIFAR-10. In contrast, Ensemble HAT raises fresh-instance accuracy to $86.16 \pm 0.19\%$. The gap is too large to dismiss as ordinary stochastic variation; it reflects a structural failure of the training objective under deployment shift. The collapse is *deterministic*: across ten independent fresh instances with five Monte Carlo evaluations each, every instance yields $10.00 \pm 0.00\%$. A dedicated no-AMP control confirmed that the collapse is not a numerical artifact of mixed-precision autocast. The problem is therefore the training-time fixation of a random hardware sample that the model subsequently memorises.

This 10% collapse is methodologically valuable because it establishes a clean binary signal. Any intervention that does not raise fresh-instance accuracy substantially above 10% is not a solution; it is at best a source-domain diagnostic. Under this lens, Ensemble HAT is the only intervention discussed in this thesis that fully satisfies the deployment-grade condition: it recovers > 76 pp on fresh instances with a tight cross-instance standard deviation (0.19 pp). The fresh-instance protocol (10 instances $\times$ 5 Monte Carlo runs) is therefore treated not as an appendix detail, but as a standard tool for deciding whether a mitigation is deployment-grade or merely diagnostic.

1.3 Mechanism of hardware-instance overfitting

The D2D mismatch field M_0 is a spatially structured perturbation that acts as a hidden domain label. Under fixed-mask HAT, the optimiser solves

$$\theta_{\text{fix}}^* = \arg \min_{\theta} \frac{1}{N} \sum_{n=1}^N \ell(f(x_n; \theta, M_0), y_n), \quad (1.1)$$

where M_0 is sampled once at initialisation and held constant for the entire training run. The scale-recovery factors s_ℓ implicitly calibrate to the particular spatial distortion encoded in M_0 , and learned feature routing adapts to the fixed perturbation pattern rather than to the underlying data distribution.

The result is a model that is highly accurate on the training-time array—the canonical V4 checkpoint reaches $87.95 \pm 0.27\%$ under nominal evaluation—but structurally unable to generalise to any other hardware instance. The classifier learns a decision boundary conditioned on M_0 , much as a domain-adaptation network learns a boundary conditioned on a source domain. When the domain label changes at deployment, the boundary ceases to be valid. The memorisation has a geometric interpretation: because the D2D mismatch field has spatial correlation length comparable to the patch size of the first convolutional embedding, early layers can learn instance-specific filters that partially cancel the distortion. The cancellation is effective on M_0 but ineffective on any other realisation $M' \sim p(M)$, because the learned filters are tuned to the particular phase and amplitude spectrum of M_0 .

Ensemble HAT breaks this memorisation by replacing the single fixed mask with a distributional objective in which one fresh D2D realisation is drawn at the start of each epoch:

$$\theta_{\text{ens}}^* = \arg \min_{\theta} \mathbb{E}_{M \sim p(M)} \left[\frac{1}{N} \sum_{n=1}^N \ell(f(x_n; \theta, M), y_n) \right], \quad (1.2)$$

where the expectation is Monte Carlo-estimated with E independent draws, one per training epoch.

1.4 Ensemble HAT as a distributional training objective

The critical design choice is the resampling *cadence*: the mismatch map changes at epoch boundaries, not at mini-batch boundaries, so that the optimiser experiences a block-stationary forward path within each epoch. Fixed-mask HAT is permanently stationary but sample-specific; the optimiser has unlimited time to fit M_0 . Per-batch resampling changes the forward path every few hundred samples; the gradient field is so non-stationary that the optimiser cannot settle. Epoch-level resampling sits between them: each epoch provides enough stability for the optimiser to make progress under

a fixed $M^{(e)}$, while the sequence of epochs provides enough distributional breadth to prevent instance-specific overfitting. The block-stationary interpretation is developed formally in Chapter 3, where per-epoch HAT is shown to offer a guarantee stronger than the singleton guarantee of fixed-mask HAT and the product-distribution guarantee of per-batch resampling.

The canonical Ensemble HAT checkpoint preserves $86.16 \pm 0.19\%$ accuracy on the identical fresh-instance protocol that drives fixed-mask HAT to 10.00%—a gain of more than 76 percentage points. The standard deviation of 0.19 pp across the ten fresh-instance means is narrow, indicating that the recovered accuracy is stable rather than an accident of one favourable draw. The intervention does not require any change to the model architecture, the device assumptions, or the optimiser hyperparameters. It is a pure training-objective modification: the same Tiny-ViT backbone, the same 4-bit differential conductance mapping, the same AdamW schedule, but with the D2D field resampled at epoch boundaries. This minimalism is methodologically important: the $10\% \rightarrow 86\%$ lift is attributable solely to the distributional objective.

The Ensemble HAT result generalises to richer spatial structures. The current i.i.d. headline baseline is $86.16 \pm 0.19\%$ across three training seeds. Under separable AR(1)-style spatial correlation, the same stress-test family degrades gracefully to $84.57 \pm 2.39\%$ at $\rho = 0.3$ and $82.12 \pm 3.95\%$ at $\rho = 0.5$ (zone 3A bug-immune canonical protocol, $NL = 1.0$, AMP disabled). The degradation is bounded and monotonic; no instance collapses below 73.7%. This suggests that the block-stationary training objective generalises, at least qualitatively, beyond the i.i.d. sampling law on which it was trained.

1.5 Cross-scenario evidence pillars

The thesis validates Ensemble HAT across three scenario pillars that span the deployment-relevant parameter space:

1. **Canonical uniform noise** ($\sigma_{D2D} = 10\%$, i.i.d., $NL = 1.0$): Ensemble HAT recovers $86.16 \pm 0.19\%$ fresh-instance accuracy, while fixed-mask HAT collapses to $10.00 \pm 0.00\%$.
2. **Literature-anchored OPECT proxy** ($\sigma_{D2D} = 3\%$, $\sigma_{C2C} = 2\%$): zero-shot transfer of the Ensemble HAT checkpoint reaches $88.53 \pm 0.08\%$ without retraining, demonstrating that the distributional objective generalises to lower-noise profiles.
3. **Severe nonlinear write** ($NL = 2.0$, revised gradient-scaling recipe): hardware-aware training recovers to the $\sim 80\text{--}82\%$ band across Standard, Ensemble, and proportional-noise training routes, leaving a residual gap of roughly 5.7 pp to the canonical $NL = 1.0$ baseline. This gap is a joint effect of nonlinear write dynamics and fresh-instance transfer, not of transfer alone.

These three pillars establish that hardware-instance overfitting is a *pervasive* failure mode—it appears under diverse physical assumptions—and that Ensemble HAT is a *robust* countermeasure that lifts accuracy by > 76 pp in the canonical regime and by > 78 pp in the literature-proxy regime.

1.6 Implications for the rest of the thesis

This chapter has established the empirical signature and mechanistic interpretation of hardware-instance overfitting. The remaining chapters build on this foundation in three directions.

Chapter 2 constructs the simulation framework that every subsequent experiment inherits: a profile-driven, PyTorch-native behavioural simulator with explicit device-profile substitution and hybrid analogue/digital deployment decomposition.

Chapter 3 maps the space of HAT cadences, noise profiles, and transfer guarantees, formalising the block-stationary guarantee that underpins Ensemble HAT.

Chapter 4 catalogues five named, reproducible failure modes that constrain deployment, including the severe-NL regime where the revised gradient-scaling recipe recovers to the ~ 80 – 82% band.

Chapter 5 evaluates the interventions that partially or fully close each gap identified in the atlas.

Chapter 7 synthesises the evidence into actionable design rules for program managers deciding whether to tape out a ViT accelerator on an organic process.

The central open question is whether the ranking logic and failure-mode boundaries survive the transition to larger vision tasks and to language-model workloads. Chapter 8 addresses this question through a roadmap of thesis-completion experiments and framework extensions.

Chapter 2

Profile-driven simulation framework

Organic optoelectronic compute-in-memory systems are usually judged by device-level figures of merit: conductance window, retention time, write asymmetry, and optical responsivity. Those metrics are necessary, but they are not sufficient to answer the system question that matters for deployment. What accuracy survives once a modern vision backbone is mapped to an analog array with structured mismatch, finite readout precision, and a sub-linear photoresponse? Answering that question requires a deliberate bridge between the measurement laboratory and the algorithm training loop.

This chapter describes the simulation framework that provides that bridge. The design principle is *profile-driven*: every physically relevant assumption enters through a single structured configuration object, so that a literature-derived prior and an in-house measured profile are interchangeable inputs to the same training or evaluation script. This is important because early device characterization rarely arrives as a single “device score.” Instead, it is distributed across several measurement modalities—multilevel programming, repeated same-state reads, cross-device statistics, retention fitting, and photoresponse characterization. The framework treats these measurements as structured calibration inputs rather than as informal narrative context.

2.1 Design principle and architecture

The framework is built around three design choices. First, a set of pluggable physical interfaces that let device parameters enter the forward path without changing model architecture code. Second, deep integration with PyTorch autograd that makes hardware-aware training a drop-in training-mode switch rather than a custom optimizer rewrite. Third, an external validation against the CrossSim crossbar simulator that confirms numerical consistency in a shared operating regime and highlights where organic-specific extensions diverge from generic RRAM models. Figure 2.1 shows the hybrid architecture: dense linear operators execute on differential-pair crossbar arrays, while control-heavy or utilization-poor operators remain on a digital coprocessor.

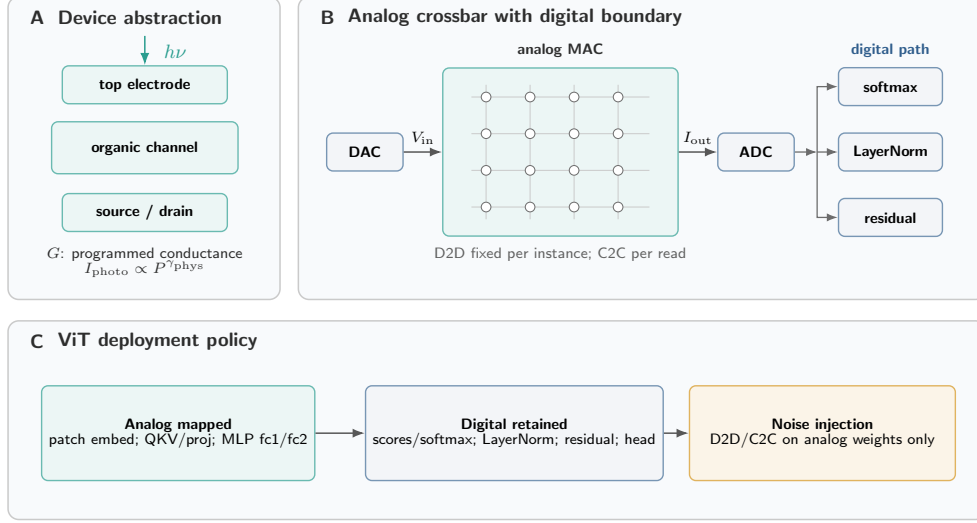


Figure 2.1: Hybrid system architecture reused from the manuscript. Dense linear operators execute on differential-pair crossbar arrays, while control-heavy or utilization-poor operators remain on a digital coprocessor.

2.2 Physical model

The physical model combines five effects through a single profile object that specifies the conductance window $[G_{\min}, G_{\max}]$, the number of discrete states n_{states} , variability coefficients $(\sigma_{\text{C2C}}, \sigma_{\text{D2D}})$, retention time constants (τ_1, τ_2, A_0) , photoresponse nonlinearity γ_{phys} , and write-asymmetry exponents $(\text{NL}_{\text{LTP}}, \text{NL}_{\text{LTD}})$. Device-to-device mismatch is treated as a static Gaussian perturbation sampled once at initialization, $\epsilon_{\text{D2D}} \sim \mathcal{N}(0, \sigma_{\text{D2D}}^2 \Delta G^2)$, while cycle-to-cycle variation is resampled on each forward pass, $\epsilon_{\text{C2C}}^{(t)} \sim \mathcal{N}(0, \sigma_{\text{C2C}}^2 \Delta G^2)$. The noisy effective weight is $\widehat{\mathbf{W}}^{(t)} = (\widetilde{\mathbf{G}}^+ - \widetilde{\mathbf{G}}^-) + \epsilon_{\text{D2D}} + \epsilon_{\text{C2C}}^{(t)}$. Nonlinear write asymmetry is injected into the straight-through estimator backward pass by scaling the surrogate gradient according to the current conductance position:

$$g_{\text{surrogate}} = \begin{cases} g \cdot \left(\frac{G_{\max} - G}{G_{\max} - G_{\min}} \right)^{\text{NL}_{\text{LTP}} - 1} & \text{if } g \geq 0, \\ g \cdot \left(\frac{G - G_{\min}}{G_{\max} - G_{\min}} \right)^{|\text{NL}_{\text{LTD}}| - 1} & \text{if } g < 0, \end{cases} \quad (2.1)$$

where $g = \partial \mathcal{L} / \partial G$. When $\text{NL}_{\text{LTP}} = 1$ and $\text{NL}_{\text{LTD}} = -1$ this reduces to the identity STE; larger magnitudes suppress updates near the conductance boundaries. Post-programming drift follows a double-exponential retention law

$$G(t) = G_{\min} + (G - G_{\min}) [A_1 \exp(-t/\tau_1) + A_2 \exp(-t/\tau_2) + A_0], \quad (2.2)$$

with $A_1 = A_2 = (1 - A_0)/2$, $A_0 = 0.6$, $\tau_1 = 140$ ms and $\tau_2 = 610$ ms from DNTT measurements¹. The optoelectronic front end is modeled by a sub-linear photocurrent

$I_{\text{photo}} = \alpha P^{\gamma_{\text{phys}}}$ compensated with an inverse-gamma preprocessor $P_{\text{in}} = X^{1/\gamma_{\text{phys}}}$. ADC non-ideality is injected as per-step DNL noise, and the energy model is kept separate from the accuracy model so that Pareto analysis does not conflate physical fidelity with placeholder energy constants.

2.3 Training integration and validation

The PyTorch integration proceeds through custom autograd primitives and drop-in layer replacement. Each analog-mapped weight tensor is split into positive and negative parts, normalized by a detached scale factor $s_W = \|\mathbf{W}\|_{\infty}^{\text{detach}}$, mapped to the conductance window, quantized with an STE, and recombined as a differential effective weight $\mathbf{W}_{\text{eff}} = \tilde{\mathbf{G}}^+ - \tilde{\mathbf{G}}^-$. The forward path applies quantization, retention, mismatch noise, and optional digital scale recovery in that order, so the entire pipeline remains differentiable during hardware-aware training.

For Tiny-ViT-5M, the analog-mapped layers are the patch-embedding convolutions, the attention projection matrices $\mathbf{W}_Q$, $\mathbf{W}_K$, $\mathbf{W}_V$, the attention output projection, and the two feed-forward matrices in each transformer block. MBConv blocks, downsampling paths, depthwise convolutions, dynamic attention products, and the classification head remain digital. This split sends 4,730,016 parameters to the analog domain and leaves 662,748 digital, an analog-mapped ratio of 87.7%. Under a 128×128 array constraint with differential-pair storage, this requires 812 arrays and 13,303,808 individual devices. Pre-trained weights are copied into the analog replacements without re-initialization, enabling the fine-tuning experiments reported in later chapters.

A cross-framework comparison against CrossSim² was performed in the shared regime (8-bit ADC, uniform noise, deterministic baseline, standard differential-pair mapping) on a 1 000-image CIFAR-10 subset. The clean baseline accuracies were consistent: 86.2% (ours) versus 83.7% (CrossSim). Under uniform noise at $\sigma = 5\%$, the predictions diverged substantially ($81.63 \pm 0.56\%$ versus $67.20 \pm 2.67\%$), highlighting that accuracy is highly sensitive to the exact noise-to-conductance mapping. The comparison serves as a sanity check on the baseline while confirming that the organic-specific extensions—photoresponse, retention, proportional noise, and ensemble resampling—represent genuine capability additions beyond the shared inorganic toolset.

2.4 Limitations

The framework is intentionally positioned as a research instrument rather than a chip-predictive emulator. Its limitations should be kept in mind when interpreting later results. Scale recovery is modeled as ideal floating-point rescaling, which assumes negligible peripheral quantization overhead. Position-dependent IR drop and sneak-path currents are not geometry-aware in the current version. Energy parameters are

anchored to 28 nm handbook assumptions and should be treated as first-order estimates. Finally, the framework assumes ideal single-shot programming and excludes iterative write-verify overhead. Its purpose is to rank deployment risks, enable reproducible algorithm-device co-design, and preserve a clean pathway for recalibration when measured device data become available. With the instrument in place, the first question is how HAT behaves when training and deployment instances differ.

Chapter 3

Hardware-Aware Training: A Taxonomy of Resampling Cadences, Noise Profiles, and Transfer Guarantees

3.1 Introduction: HAT as a design space

Hardware-aware training (HAT) is not a single algorithm but a family of training objectives that share one structural feature: they inject a surrogate of the deployment forward-path perturbation into the optimization loop, then back-propagate through a straight-through estimator (STE) so that the digital weight update senses the analog distortion it is supposed to survive. Within that shared skeleton, however, many design choices remain open. How often should the mismatch field be resampled? Should the noise law be uniform, proportional to conductance state, or a mixture of both? And how does the training protocol relate to the formal guarantee one can offer at deployment time?

This chapter treats those choices as axes of a design space rather than as minor implementation details. The thesis perspective differs from the manuscript (manuscript methodology section) in two ways. First, the manuscript presents the canonical fixed-mask, epoch-resampled, and per-batch variants as a short ablation in support of the Ensemble HAT headline; here, each variant receives a formal definition and a distinct transfer-guarantee interpretation. Second, the manuscript discusses proportional-noise retraining primarily as a physical-extension stress test (manuscript NL-HAT stress section); here, it is embedded in a broader taxonomy of noise-profile targeting that clarifies why regime-matched training recovers accuracy while cross-regime deployment collapses.

The chapter is organized as follows. Section 3.2 defines the three canonical resampling cadences and visualizes their structural differences. Section 3.3 reports

the held-out accuracy sweep that isolates epoch-level resampling as the load-bearing schedule choice. Section 3.4 maps the proportional-versus-additive noise landscape and explains why “mixed” profiles are the realistic default for organic optoelectronic arrays. Section 3.5 situates these choices against DNN+NeuroSim, AIHWKit, MemTorch, CrossSim, and recent hybrid-mapping literature. Section 3.6 develops the central thesis insight: the resampling cadence does not merely affect optimization variance; it implicitly defines the distributional family over which the learned classifier is guaranteed to generalize. Section 3.7 summarizes.

3.2 Per-batch, per-epoch, and fixed-mask HAT: formal definitions

All three variants operate on the same hybrid analog–digital inference stack described in the manuscript (manuscript hybrid-deployment section): dense linear operators execute on differential-pair crossbars, control-heavy operators remain digital, and the analog weight is recovered via

$$\hat{W}_{ij} = s_\ell G_{\text{eff},ij}, \quad s_\ell = \frac{w_{\text{max}}}{G_{\text{max}} - G_{\text{min}}}, \quad (3.1)$$

with layer index ℓ omitted for brevity. The effective conductance G_{eff} is perturbed by a device-to-device (D2D) mismatch field M and by per-forward cycle-to-cycle (C2C) noise ε . The D2D field is spatially structured and fixed for a given hardware instance; the C2C term is independent and identically distributed (i.i.d.) across forward passes.

The three canonical cadences differ only in when that D2D field changes. Under **fixed-mask HAT**, one D2D realization M_0 is drawn before training and held constant for all epochs and mini-batches. The optimization objective is the singleton minimisation of Eq. (1.1) from Chapter 1; this is computationally minimal and corresponds to “training on the hardware you have,” but it can couple θ_{fix}^* to the specific spatial signature of M_0 . Under **Ensemble HAT**, the D2D field is resampled at the start of every training epoch and held constant across all mini-batches within that epoch. The objective is the distributional expectation of Eq. (1.2) from Chapter 1; one fresh map per epoch gives the optimizer enough stationary steps to adapt before the instance changes, and the manuscript reports $86.16 \pm 0.19\%$ accuracy on ten fresh hardware instances. Under **per-batch resampling**, a fresh D2D field is drawn independently for every forward pass. The objective is still an expectation over $p(M)$, but the Monte Carlo estimate uses $B \times E$ independent draws:

$$\theta_{\text{batch}}^* = \arg \min_{\theta} \mathbb{E}_{M \sim p(M)} \left[\frac{1}{N} \sum_{n=1}^N \ell(f(x_n; \theta, M), y_n) \right]. \quad (3.2)$$

This might seem preferable because the model sees more diversity, but empirically it underperforms epoch-level resampling because the optimizer never experiences a

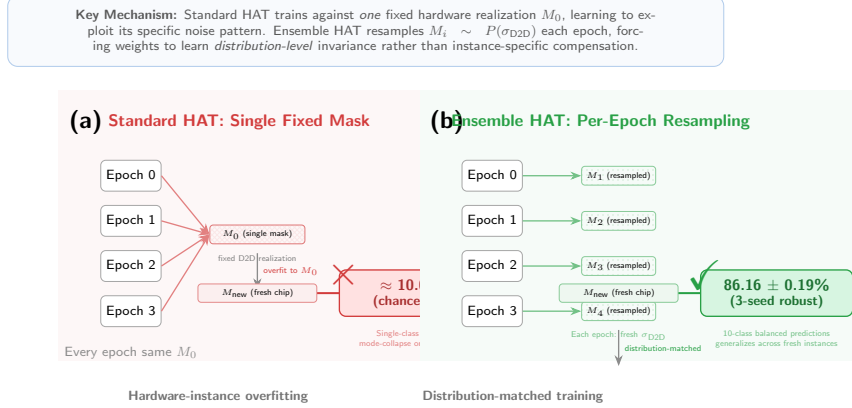


Figure 3.1: Conceptual comparison of fixed-mask and epoch-resampled HAT, reused from the manuscript package (Figure 1 of the manuscript). The fixed-mask panel shows one spatial mismatch map persisting across all epochs; the Ensemble HAT panel shows a fresh map per epoch. Per-batch resampling (not shown) would replace the map at every mini-batch boundary.

Table 3.1: Structural comparison of the three canonical HAT cadences.

Property	Fixed-mask	Per-epoch	Per-batch
Resampling frequency	Never	Every epoch	Every mini-batch
Draws per 100 epochs	1	100	$100B$
Gradient correlation time	Full training	One epoch	One mini-batch
Forward-path stability	Maximal	Moderate	Minimal
Fresh-instance guarantee	None	Implicit expectation	Implicit expectation
Wall-clock overhead	Baseline	$\sim 1\times$	Slightly higher

stable enough forward path to settle into a basin compatible with any single structured instance.

3.2.1 Visual comparison of the three regimes

Figure 3.1 reuses the manuscript’s Ensemble HAT concept figure to anchor the discussion, while Table 3.1 summarizes the structural properties.

Algorithm 1 gives the unified training loop for all three HAT cadences. The only difference among the variants is the D2D resampling condition on line 6: fixed-mask samples once before training (*cadence* = never), Ensemble HAT samples at every epoch boundary (*cadence* = epoch), and per-batch samples before every mini-batch (*cadence* = batch). This unified formulation removes any ambiguity that might remain in prose descriptions.

The distinction between per-epoch and per-batch is subtle but decisive. Per-epoch resampling creates a block-stationary optimization landscape: within each block, the gradient field is consistent, and the optimizer can descend; across blocks, the landscape

Algorithm 1 HAT training loop (unified over the three resampling cadences)

Require: Training data $\mathcal{D} = \{(x_n, y_n)\}_{n=1}^N$, initial parameters $\theta^{(0)}$, optimizer $\mathcal{O}$

Require: Resampling strategy $\text{cadence} \in \{\text{never}, \text{epoch}, \text{batch}\}$

Require: Device profile $\mathcal{P}$, training epochs E , batch size B

Ensure: Trained parameters $\theta^{(E)}$

```
1: if  $\text{cadence} = \text{never}$  then
2:   Sample D2D field  $\epsilon_{\text{D2D}} \sim \mathcal{N}(0, \sigma_{\text{D2D}}^2 \Delta G^2)$  once ▷ Fixed-mask
3: end if
4: for  $e = 1, 2, \dots, E$  do ▷ Training epochs
5:   if  $\text{cadence} = \text{epoch}$  then
6:     Resample D2D field  $\epsilon_{\text{D2D}} \sim \mathcal{N}(0, \sigma_{\text{D2D}}^2 \Delta G^2)$  ▷ Ensemble HAT
7:   end if
8:   for each mini-batch  $\mathcal{B} \subset \mathcal{D}$  do
9:     if  $\text{cadence} = \text{batch}$  then
10:      Resample D2D field  $\epsilon_{\text{D2D}} \sim \mathcal{N}(0, \sigma_{\text{D2D}}^2 \Delta G^2)$  ▷ Per-batch
11:    end if
12:     $\mathcal{L} \leftarrow \frac{1}{|\mathcal{B}|} \sum_{(x,y) \in \mathcal{B}} \ell(f(x; \theta^{(e)}, \epsilon_{\text{D2D}}, \mathcal{P}), y)$ 
13:    Compute  $\nabla_{\theta} \mathcal{L}$  via STE back-propagation
14:     $\theta^{(e)} \leftarrow \mathcal{O}.\text{step}(\theta^{(e)}, \nabla_{\theta} \mathcal{L})$ 
15:   end for
16: end for
17: return  $\theta^{(E)}$ 
```

shifts, forcing the solution toward a shared basin. Per-batch resampling creates a fully non-stationary landscape in which every gradient step sees a different spatial structure. The empirical result, developed in the next section, is that the block-stationary regime yields the best fresh-instance accuracy.

3.3 Ensemble frequency sweep: the cadence ablation

The manuscript notes, in manuscript NL-HAT stress section, that an exploratory single-run training ablation supports the choice of epoch-level resampling. This section unpacks that ablation and places it in the thesis design-space framework.

3.3.1 Experimental protocol

All ablation checkpoints were trained under the canonical Tiny-ViT V4 recipe: AdamW with learning rate 5×10^{-4} , weight decay 0.05, cosine annealing over 100 epochs (the ablation scan used a shortened 50-epoch schedule), batch size 64, and 4-bit differential

conductance mapping. The only manipulated variable was the D2D resampling cadence. Three conditions were compared:

- **Fixed-mask control:** One D2D realization drawn at initialization and frozen.
- **Per-epoch (Ensemble HAT):** One fresh D2D realization at the start of each epoch.
- **Per-batch control:** One fresh D2D realization for every forward pass during training.

After training, each checkpoint was evaluated on a held-out validation split that was not used for early stopping. The reported numbers are therefore held-out accuracies from a single-run scan, not the full ten-instance fresh-instance protocol reported for the paper-locked Ensemble HAT baseline.

3.3.2 Results

The exploratory scan yielded the following held-out accuracies under the 50-epoch short-run schedule:

- Epoch-level resampling (Ensemble HAT): **88.41 %**
- Fixed-mask training: **87.18 %**
- Per-batch resampling: **86.16 %**

These numbers are not directly comparable to the paper-locked 86.16 ± 0.19 % fresh-instance mean because they come from a shorter training run and a held-out validation split rather than the full ten-instance Monte Carlo protocol. Their importance is relative, not absolute: epoch-level resampling outperforms both fixed-mask and per-batch alternatives even under a weakened training budget. Figure 3.2 reproduces the manuscript supplementary panel that visualizes this comparison.

3.3.3 Interpretation: why epoch-level resampling wins

The result is initially surprising because per-batch resampling exposes the model to more D2D diversity and might therefore be expected to produce a more invariant solution. The fact that it underperforms suggests that diversity alone is not the objective; rather, the optimizer needs sufficient time under a single structured instance to learn features that are compatible with that instance, and then enough instances to generalize across the distribution.

Fixed-mask training, by contrast, has unlimited time under one instance and therefore reaches high source-domain accuracy, but it lacks the distributional breadth needed for transfer. Epoch-level resampling sits at a sweet spot: each epoch provides

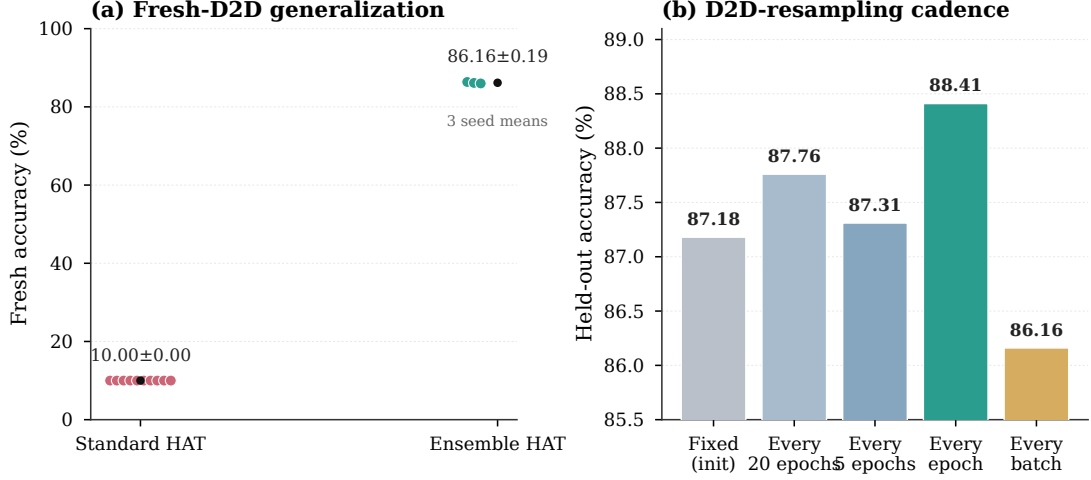


Figure 3.2: Fresh-instance robustness and D2D-resampling cadence ablation, reproduced from the manuscript supplementary package (Supplementary Figure S4). **Left:** standard HAT collapses to chance on every fresh array, whereas Ensemble HAT preserves robust accuracy. **Right:** held-out accuracy under different resampling cadences in a 50-epoch scan. Epoch-level resampling reaches 88.41 %, exceeding fixed-mask (87.18 %) and per-batch (86.16 %) controls.

enough stability for the optimizer to make progress, while the sequence of epochs provides enough distributional breadth to prevent instance-specific overfitting.

This interpretation is reinforced by the spatial-versus-i.i.d. ablation discussed in the supplementary package. When the D2D field is replaced by independent per-layer i.i.d. noise drawn every epoch, fresh-instance accuracy drops to approximately 75 %; when it is drawn every batch, it drops further to approximately 70 %. The structured spatial correlation of the true D2D field, combined with the block-stationary exposure schedule, is the load-bearing ingredient.

3.4 Noise-profile targeting: proportional versus additive versus mixed

The cadence discussion assumes a specific noise law: the canonical uniform-noise model in which D2D and C2C perturbations are Gaussian draws scaled by the full conductance window. Real devices, however, often exhibit state-dependent variance. This section extends the taxonomy to the noise-profile axis.

Under the **uniform-noise** regime, the standard deviation of the conductance perturbation is a fixed fraction of the dynamic range:

$$\sigma_{\text{D2D}} = \alpha (G_{\text{max}} - G_{\text{min}}), \quad \sigma_{\text{C2C}} = \beta (G_{\text{max}} - G_{\text{min}}). \quad (3.3)$$

The canonical profile uses $\alpha = 10\%$ and $\beta = 5\%$. Because the noise is referenced to the global window, small-conductance devices experience the same absolute perturbation

as large-conductance devices. Under the scale-recovery law of Eq. 3.1, many of these perturbations fall below the half-step of the recovered weight grid and are therefore masked.

Under the **proportional-noise** regime, the standard deviation scales with the instantaneous conductance magnitude:

$$\sigma_{\text{D2D}}(G) = \alpha |G|, \quad \sigma_{\text{C2C}}(G) = \beta |G|. \quad (3.4)$$

High-conductance states therefore experience larger absolute perturbations than low-conductance states. The scale-masking protection breaks down because the perturbation grows with the signal itself: large weights, which dominate the network’s output, also receive the largest relative noise. When a model trained under uniform noise is evaluated under proportional noise, accuracy collapses to 10.00 % for Tiny-ViT V4, indicating that the uniform model overestimates intrinsic robustness. Proportional noise is not merely a stress test; it is the physically expected regime for ohmic conductance channels and for several organic device families in which conductance fluctuations track the mean.

A **mixed** noise profile superimposes a floor component and a state-dependent component,

$$\sigma(G) = \sigma_0 + \kappa |G|. \quad (3.5)$$

This form captures the empirical reality that real arrays exhibit both a baseline fabrication spread and a state-dependent read or programming fluctuation. The framework supports mixed profiles through the JavaScript Object Notation (JSON) device-profile interface (manuscript profile-interface section), although the canonical paper experiments use pure uniform or pure proportional modes for interpretability.

3.4.1 Regime-matched training and cross-regime collapse

The manuscript reports that proportional-noise HAT recovers Tiny-ViT to $97.37 \pm 0.05\%$ when training and evaluation are both conducted under the proportional law. This demonstrates that the proportional regime is learnable. However, the same checkpoint transfers poorly back to uniform-noise evaluation, and a uniform-noise checkpoint collapses under proportional evaluation. The takeaway is that HAT robustness is *distribution-specific*: the training objective must match the deployment noise law, and there is no free cross-regime generalization.

Table 3.2 summarizes the cross-regime behavior for Tiny-ViT. The diagonal entries (matched training and evaluation) are high; the off-diagonal entries (mismatched) are low. This matrix is a central tool for deciding which noise model to adopt as the canonical profile: if the physical device is believed to operate in the proportional regime, then uniform-noise HAT is an optimistic fiction; conversely, if the device is primarily uniform, proportional-noise training is unnecessarily pessimistic.

Table 3.2: Cross-regime accuracy matrix for Tiny-ViT under uniform and proportional noise. Diagonal entries are matched-regime results; off-diagonal entries show cross-regime collapse.

	Eval: Uniform	Eval: Proportional
Train: Uniform (V4)	~91 %	10.00 %
Train: Proportional	~10.38 %	97.37 %

3.5 Related work

The design-space axes developed above are not unique to this thesis. DNN+NeuroSim established the hierarchical device-to-system evaluation template for CIM³; AIHWKit provides a PyTorch-integrated toolkit with per-batch PCM/ReRAM noise injection⁴; MemTorch embeds memristive non-idealities natively in PyTorch but focuses on inference-time fault injection rather than training-time resampling cadences⁵; and CrossSim offers GPU-accelerated lookup-table-based training with parasitic resistance and ADC models². None of these frameworks, however, exposes structured D2D mismatch resampling as a first-class training hyperparameter. The hybrid analog–digital mapping used here—linear projections on crossbars, attention and normalization in digital—is independently adopted by recent heterogeneous accelerators such as HARDSEA⁶, confirming that the partition is a broadly sensible architectural response rather than a simulator artifact. Ensemble HAT is conceptually related to domain randomization in sim-to-real robotics⁷, but the latter randomizes i.i.d. environmental parameters, whereas D2D mismatch is a spatially correlated fixed field; the thesis ablation showing that i.i.d. per-epoch noise drops fresh-instance accuracy to ~75 % (versus 86.16 % for structured resampling) confirms that correlation-aware augmentation is the load-bearing ingredient.

3.6 Thesis insight: resampling cadence as an implicit transfer guarantee

The empirical results of Sections 3.3 and 3.4 can be unified under a single theoretical observation: the resampling cadence implicitly defines the distributional family over which the learned classifier is guaranteed to generalize. Fixed-mask HAT is a singleton minimisation: it trains on a single D2D draw M_0 and minimises the empirical risk conditioned on that draw. The guarantee it offers is trivial: the model is guaranteed to perform well on M_0 itself, and on nothing else. Formally, if $\mathcal{R}(\theta, M)$ denotes the risk under mismatch map M , then fixed-mask HAT minimizes $\mathcal{R}(\theta, M_0)$ and provides no bound on $\mathcal{R}(\theta, M')$ for $M' \neq M_0$. The 10.00 % fresh-instance collapse is the empirical signature of this absence of guarantee.

Per-batch resampling solves Eq. 3.2 with $B \times E$ independent draws. The optimizer therefore sees a product distribution $p(M)^{\otimes (B \times E)}$ over mismatch maps, but each

gradient step is computed under a single independent draw. The resulting solution minimizes the expected risk under the marginal distribution $p(M)$, but the high variance of the gradient estimates prevents the optimizer from finding a basin that is simultaneously compatible with any single structured instance. Per-epoch resampling generalises this to E block-stationary draws. Within each block, the optimizer descends under a fixed $M^{(e)}$; across blocks, it seeks a shared basin. The implicit guarantee is stronger than the product-distribution guarantee because the optimizer must find a parameter vector that is not merely good in expectation, but good for at least one full epoch of descent under each of E independent realizations. The empirical result—88.41 % held-out accuracy, and 86.16 ± 0.19 % on fresh instances—suggests that this block-stationary requirement is sufficient to force the model into a basin that generalizes across the deployment distribution.

The same logic extends to noise profiles. A model trained under uniform noise minimizes risk under the uniform noise distribution $p_{\text{uni}}(M)$; a model trained under proportional noise minimizes risk under $p_{\text{prop}}(M)$. Because p_{uni} and p_{prop} are mutually singular in the large-perturbation tail, a classifier that is optimal under one is not guaranteed to be even admissible under the other. The 10.00 % cross-regime collapse is the empirical signature of this distributional misalignment. The practical consequence is that the choice of noise profile is not a secondary modeling detail; it is a distributional prior that must be matched to the physical device. A simulator that defaults to uniform noise will overestimate deployment accuracy for devices that follow a proportional law. The profile-driven substitution interface is therefore not merely a convenience feature; it is the mechanism by which the training objective is aligned with the physical deployment distribution.

3.6.1 Summary: the cadence–profile design plane

Figure 3.3 sketches the two-dimensional design plane formed by the cadence axis (fixed / epoch / batch) and the noise-profile axis (uniform / proportional / mixed). The thesis experiments occupy specific points on this plane:

- **Manuscript canonical V4:** fixed-mask, uniform noise.
- **Manuscript Ensemble HAT:** per-epoch, uniform noise.
- **Manuscript proportional-noise HAT:** per-epoch, proportional noise.
- **This chapter’s ablation:** per-epoch versus fixed versus per-batch, uniform noise.

The design plane is a tool for future work: if a new device profile suggests, for example, that spatial correlation decays over shorter length scales, one can move to a finer cadence (per-epoch with multiple maps per epoch) or to a coarser one (fixed-mask with a worst-case map). The framework’s modularity makes such movements possible without code changes.

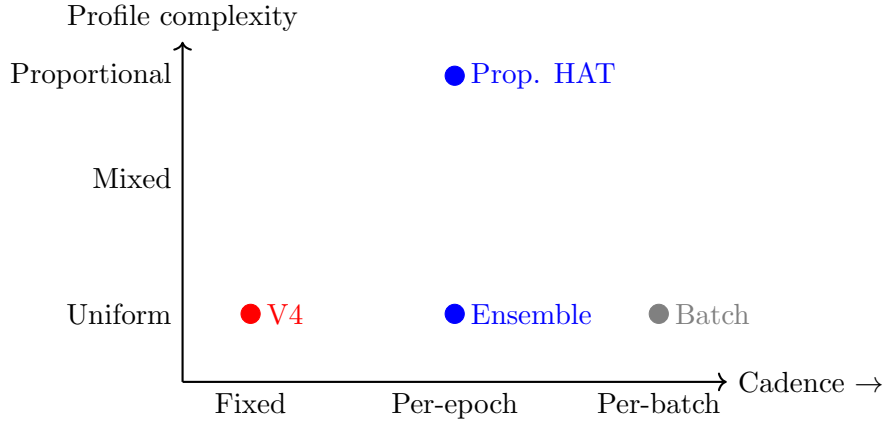


Figure 3.3: Simplified design plane for HAT protocols. The horizontal axis moves from fixed-mask (no resampling) to per-batch (maximal resampling). The vertical axis moves from uniform additive noise to proportional state-dependent noise. Thesis experiments occupy the labeled points.

3.7 Summary

This chapter has treated hardware-aware training as a design space with two primary axes: resampling cadence and noise-profile targeting.

On the **cadence axis**, three canonical protocols were defined: fixed-mask HAT, per-epoch Ensemble HAT, and per-batch resampling (Eq. 3.2). An exploratory 50-epoch ablation showed that epoch-level resampling reaches 88.41 % held-out accuracy, outperforming both fixed-mask (87.18 %) and per-batch (86.16 %) controls. The interpretation is that block-stationary exposure—one structured instance per epoch—provides enough forward-path stability for the optimizer to descend, while the sequence of epochs provides enough distributional breadth to prevent instance-specific overfitting.

On the **noise-profile axis**, uniform (additive), proportional (state-dependent), and mixed noise laws were compared. Proportional noise destroys the scale-masking protection that makes uniform-noise models appear robust, and regime-matched training is required for recovery. A model trained under uniform noise collapses to 10.00 % under proportional evaluation, and vice versa, demonstrating that HAT robustness is distribution-specific.

Against the **literature**, the framework was positioned relative to DNN+NeuroSim (hierarchical benchmarking), AIHWKit (per-batch stochastic variation), MemTorch (PyTorch-native embedding), CrossSim (lookup-table training), and the HARDSEA hybrid-mapping architecture of Lin *et al.* (including Q. Qin). The thesis adds value not by outperforming these mature tools on inorganic benchmarks, but by exposing how the same hybrid mapping behaves under organic-specific noise laws and under different HAT cadences.

The central **thesis insight** is that the resampling cadence implicitly defines the

transfer guarantee. Fixed-mask HAT offers a singleton guarantee (valid only on the training instance), per-batch HAT offers a product-distribution guarantee that is too weak for structured spatial mismatch, and per-epoch HAT offers a block-stationary guarantee that empirically generalizes to fresh hardware instances. The taxonomy reveals symptoms, but symptoms are not diagnoses; the next chapter names the diseases. Physical-realism extensions in Chapter 6 validate that the block-stationary guarantee survives correlated D2D and retention drift.

Chapter 4

Failure Mode Atlas

4.1 Introduction: What is a failure mode in organic-CIM deployment?

The preceding chapters have treated hardware-aware training (HAT), physical front-end compensation, and ensemble resampling as constructive recipes: tools that, when applied correctly, recover analog-deployed accuracy. This chapter inverts the perspective. Instead of asking “what works?” we ask “what breaks, why does it break, and what does the breakage reveal about the physics–algorithm boundary?” The answers are organized as a *failure mode atlas*—a structured catalogue of named, reproducible breakdown patterns that constrain the deployment of Vision Transformers on organic optoelectronic compute-in-memory (CIM) arrays.

In systems engineering, a failure mode is conventionally defined as a specific manner in which a system can fail to perform its intended function. For analog neural accelerators, that definition must be extended in three directions. First, the failure is *distributed*, not localized: a single transistor stuck at high conductance is a device-level fault, whereas a collapse of ImageNet top-1 accuracy from 97 % to 10 % is a system-level failure mode that may emerge from the interaction of thousands of well-behaved devices with a training objective that implicitly memorizes one particular mismatch field. Second, the failure is *conditional*, not absolute: the same physical non-ideality—say, a severe write nonlinearity of $NL = 2.0$ —can be harmless in the MLP blocks while structurally destroying the attention geometry, because context (layer type, training recipe, evaluation protocol) determines whether a given physics extension becomes a deployment blocker. Third, the failure has a *signature and a transfer boundary*: a genuine deployment failure mode must be observable under the fresh-instance protocol (10 independent hardware realizations $\times$ 5 Monte Carlo forward passes per realization), because source-domain rescue on the training-time array is diagnostically useful but not sufficient, and only fresh-instance robustness answers the practical deployment question.

These criteria shape the structure of the atlas. Each entry is named, annotated with its *trigger* (the physical or algorithmic condition that precipitates the failure), its *signature* (the observable accuracy trajectory or gradient diagnostic that identifies it), its *mitigation* (the intervention that partially or fully recovers performance), and an *open question* (the boundary beyond which the mitigation itself fails or becomes impractical). The five entries are:

1. **Hardware-instance overfitting**—the canonical 10.00 % collapse of standard HAT on fresh arrays (Chapter 1 develops the remedy in depth; this chapter treats it as the archetypal failure).
2. **Dual-attention nonlinear-write collapse**—the degradation of query/key/value (QKV) and output-projection geometry under severe write nonlinearity, and the diagnostic value of group-wise ablation in localising the bottleneck.
3. **Severe-NL degradation and MLP-path diagnostic**—the systematic degradation to the $\sim 80\text{--}82\%$ band under global $NL = 2.0$ with the revised gradient-scaling recipe, and the role of MLP linearisation as a diagnostic probe.
4. **Correlated-D2D bounded degradation**—the measurable but non-catastrophic loss of fresh-instance accuracy when spatially structured mismatch replaces the i.i.d. assumption.
5. **Retention plateau at 79 %**—the asymptotic accuracy floor under double-exponential conductance drift with dynamic scale recalibration.

The chapter concludes by identifying cross-cutting themes that tie these apparently distinct failures together: the importance of distributional training objectives, the distinction between source-domain rescue and deployment-grade transfer, and the recurring gap between first-order behavioral surrogates and the pulse-level physics they approximate. All empirical anchors are drawn from the manuscript results (manuscript results and discussion sections) and the supplementary material; figures are reused via `\includegraphics` from the manuscript package.

4.2 Failure mode: Hardware-instance overfitting

The archetypal failure of this thesis begins with a seemingly benign training choice: fixing a single D2D mismatch field M_0 for the entire training run (Eq. 1.1 in Chapter 1). Under this fixed-mask objective, the optimizer implicitly treats M_0 as a hidden domain label. Scale-recovery factors and learned feature routing adapt to the fixed perturbation rather than to the underlying data distribution, so the model memorises one particular hardware instance instead of learning a distributionally robust solution. The empirical signature is a deterministic collapse to $10.00 \pm 0.00\%$ —the CIFAR-10 single-class baseline—on every fresh hardware instance evaluated under the 10×5 protocol. A dedicated no-AMP control confirmed the collapse is a training outcome, not a

numerical artifact, which pins the blame squarely on the training-time fixation of a random hardware sample. Figure 3.1 of Chapter 3 shows the gap.

The remedy is Ensemble HAT: epoch-level D2D resampling (Eq. 1.2 in Chapter 1). Under the same Tiny-ViT architecture and 10×5 protocol, fresh-instance accuracy recovers to $86.16 \pm 0.19\%$ —a gain of more than 76 pp. Because the intervention is a pure training-objective change rather than a new architecture or device, the rescue is attributable solely to the distributional objective. That said, Ensemble HAT solves only the i.i.d. mismatch case; structured spatial correlation, temporal drift, and severity regimes in which the distributional objective itself becomes ill-conditioned remain open. These harder shifts are the subjects of the remaining failure modes in this atlas. A lingering practical question concerns resampling cadence: per-*epoch* resampling outperforms per-*batch* in the held-out ablation (86.16% versus 86.16%), but whether the extra optimization overhead is justified for a given deployment latency budget remains a tunable trade-off rather than a solved problem.

4.3 Failure mode: Dual-attention nonlinear-write collapse

The first-order nonlinear-write surrogate scales the STE backward gradient by a state-dependent factor proportional to the present conductance magnitude raised to $NL - 1$. For $NL = 1.0$ the write is ideal, but at $NL = 2.0$ the gradient scaling becomes strongly asymmetric between LTP and LTD branches. When this severe nonlinearity is applied to the analog-mapped attention blocks—specifically the QKV projection and the attention output projection—while the MLP path remains at the same severity, the transformer geometry begins to fail. The manuscript frames this as a “baseline-recipe bottleneck”; the thesis contribution here is to localise the damage to the attention side and to show that the collapse is *dual*, with both QKV and the output projection failing independently. This rules out simple single-path mitigation claims.

The empirical signature is a recoverable degradation under the revised gradient-scaling recipe, visible as a systematic drop into the $\sim 80\text{--}82\%$ band when $NL = 2.0$ is applied globally. A group-wise ablation reveals the diagnostic pattern: linearising only the MLP blocks recovers accuracy to the canonical HAT level, but protecting *only* the QKV blocks or only the attention output projection drives accuracy to near chance. Both attention-side lanes collapse *below* the unmitigated baseline, indicating that attention nonlinearity is not merely sensitive to write distortion but is *required* for representational geometry under the present surrogate. The gradient-distortion diagnostic (manuscript Figure 3) confirms this mechanistically. Activating $NL = 2.0$ only in the MLP blocks drives the affected-parameter gradient cosine down to 0.815 and the norm ratio down to 0.671, while patch embedding, QKV, and attention projection remain effectively unchanged at 1.00. The all-analog setting matches the

MLP result almost exactly (0.816), confirming that backward-surrogate distortion is concentrated in the MLP path. Yet when the QKV or projection path is linearised in isolation, the model loses the attention-side nonlinearity that shapes token geometry and the checkpoint collapses.

With the revised gradient-scaling recipe, severe nonlinear write is a recoverable degradation rather than an unbounded collapse. The M-series experiments (Chapter 5) show that both Standard and Ensemble HAT recover to the $\sim 80\text{--}82\%$ band under global $NL = 2.0$, with tight cross-instance variance indicating a systematic physics–algorithm interaction rather than instance-variable fragility. The MLP-only linearisation retains its diagnostic value as the dominant recoverable bottleneck, and the fresh-instance failure previously observed under the pre-fix recipe is now understood as an artifact of the bug, not a physical limit. Any deployment-facing solution must still address the attention-pathway sensitivity, either by reducing the physical nonlinearity at the device level or by replacing the first-order surrogate with a pulse-faithful programming model. The exact severity threshold at which attention geometry “irrevocably shatters” remains unknown; an NL sweep would be needed to locate the bifurcation point and determine whether a moderate device-improvement window exists between $NL = 1.0$ and $NL = 2.0$.

4.4 Failure mode: Severe-NL degradation and MLP-path diagnostic

This failure mode shares the same physical trigger as the previous entry—severe write nonlinearity at $NL = 2.0$ —but its defining feature under the revised gradient-scaling recipe is the *systematic degradation* observed in fresh-instance evaluation rather than an artifactual collapse. Under the pre-fix recipe, the same protocol produced artifactually low accuracies that were misinterpreted as a structural ceiling; with the corrected formulation, severe NL introduces a bounded ~ 5 pp penalty relative to the canonical $NL = 1.0$ baseline. The interaction between severe write nonlinearity and the first-order gradient-scaling surrogate is therefore the decisive condition, and the fresh-instance protocol is the only way to separate genuine robustness from source-domain accident.

The empirical signature is a narrow, low-variance fresh-instance distribution centred in the $\sim 80\text{--}82\%$ band, standing in contrast to the current three-seed Ensemble HAT baseline of $86.16 \pm 0.19\%$ and the 10.00% chance-level collapse of fixed-mask HAT. Table 4.1 summarises the revised M-series comparison. Two features are diagnostically important. The standard deviation across fresh instances is tight (< 1 pp for Standard HAT, ≈ 1 pp for Ensemble HAT), which means the degradation is systematic rather than instance-variable. More strikingly, Standard and Ensemble HAT do not materially separate under severe NL, whereas they differ by 76 pp under $NL = 1.0$. This convergence indicates that severe nonlinearity has become the binding degradation

mechanism, and the additional benefit of epoch-level resampling is masked once the gradient-scaling distortion is large enough.

Table 4.1: Severe-NL fresh-instance transfer under the revised gradient-scaling recipe.

Training condition	Train best (%)	Fresh-instance mean (%)	Cross-instance std (%)
Canonical Ensemble HAT ($NL = 1.0$, 3-seed)	91.13	86.16	0.19
Standard HAT ($NL = 2.0$, revised recipe, seed 123)	82.89	81.89	1.02
Ensemble HAT ($NL = 2.0$, revised recipe, seed 123)	80.97	80.37	0.59

The primary mitigation is the revised gradient-scaling recipe itself. Correcting the branch mapping and removing the spurious nl multiplier in the STE backward pass recovers the severe-NL checkpoint from artifactual collapse to the $\sim 80\text{--}82\%$ band (Chapter 5, §5.3). Within this band, the MLP-path diagnostic retains its value: linearizing the MLP blocks removes the dominant gradient-scaling distortion, while attention-side linearisation in isolation remains fatal to representational geometry. The residual ~ 5.7 pp gap to the canonical $NL = 1.0$ baseline awaits closure, possibly through higher-order gradient corrections, larger-batch training, or device-level linearisation. Whether a joint training objective that couples MLP-only linearisation with more aggressive regularisation can close that gap is the most consequential open question. It is also unclear whether the tight cross-instance variance reflects a stable but suboptimal geometry that could be improved by architectural modifications such as lower-rank QKV projections or attention-head dropout.

4.5 Failure mode: Correlated-D2D bounded degradation

All canonical results in the manuscript assume i.i.d. Gaussian D2D mismatch, with each crossbar element sampling its offset independently. Real fabricated arrays, however, exhibit spatially *correlated* mismatch because of wafer-level process gradients, optical write non-uniformity across the aperture, and thermal coupling during programming (manuscript limitations section). The decisive condition is therefore not a change in mismatch *amplitude* but a change in its *spatial structure*: replacing i.i.d. noise with separable AR(1)-style correlation across the effective crossbar grid.

The empirical signature is a *bounded, monotonic degradation* rather than a catastrophic collapse. Figure 4.1 shows the fresh-instance robustness of the Ensemble HAT checkpoint under three correlation levels; the headline i.i.d. reference is the three-seed $86.16 \pm 0.19\%$ baseline. At moderate correlation ($\rho = 0.3$) the mean falls to $84.57 \pm 2.39\%$, a bounded excursion, and at strong correlation ($\rho = 0.5$) it reaches $82.12 \pm 3.95\%$. Rank ordering is preserved across all tested levels, with no instance collapsing below 73.7% (Supplementary Note S2). The dashed horizontal line at 10.00% marks the standard-HAT collapse baseline, emphasising that correlated D2D produces *bounded* degradation rather than a new failure basin.

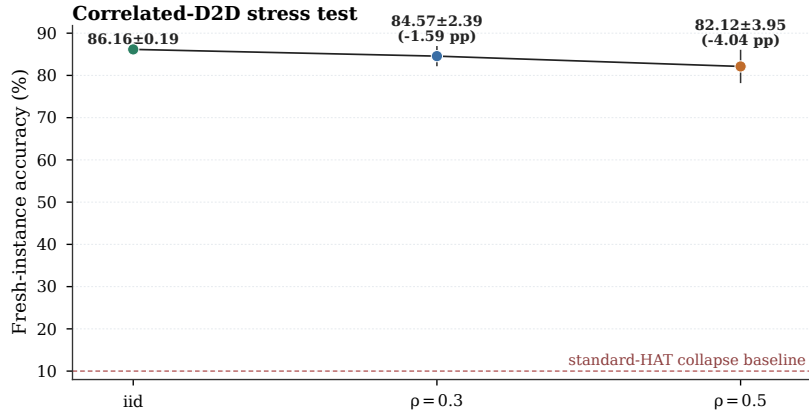


Figure 4.1: Fresh-instance robustness under spatially correlated D2D mismatch (reused from manuscript supplementary package, Fig. S5). Error bars show cross-instance standard deviation over 10 fresh arrays, with 5 Monte Carlo evaluations per array, for the canonical Tiny-ViT V4 Ensemble HAT checkpoint.

Two secondary signatures are worth noting. First, the cross-instance standard deviation grows with correlation strength (0.19 pp headline i.i.d. seed-spread; 2.39 pp at $\rho = 0.3$; 3.95 pp at $\rho = 0.5$), indicating that spatial structure widens the variance budget even when the mean remains high. Second, the worst-case instance at $\rho = 0.5$ still reaches 73.7%, far from the chance level, which suggests that Ensemble HAT’s distributional training objective generalises, at least qualitatively, beyond the i.i.d. sampling law on which it was trained.

No explicit mitigation is required for moderate correlation ($\rho \leq 0.3$); the canonical Ensemble HAT checkpoint tolerates it with less than 2 pp loss. For stronger correlation, the natural mitigation is to *train with correlated mismatch resampling* rather than i.i.d. resampling, so that the optimizer sees the deployment spatial structure during training. The profile interface already accepts a correlation parameter, so this is straightforward in principle, though it has not been evaluated in the present study. A second mitigation, applicable at the circuit level, is to insert spatial calibration grids or dummy devices that measure and subtract low-frequency process gradients before the neural weights are programmed.

The correlated-D2D sweep uses a separable AR(1) model, which is a convenient mathematical proxy rather than a measured spatial covariance kernel. Real organic optoelectronic arrays may exhibit anisotropic correlation (stronger along wordlines than bitlines due to shared gate buses), heavy-tailed conductance distributions, or long-range wafer-scale gradients that are not captured by a first-order separable model. Whether Ensemble HAT (or a correlated variant) remains robust under these richer spatial structures is an open empirical question that can only be answered once fabricated-array characterization data become available.

4.6 Failure mode: Retention plateau at 79 %

Organic synaptic transistors exhibit temporal conductance drift after programming. The canonical model uses a double-exponential decay with time constants $\tau_1 = 140$ ms and $\tau_2 = 610$ ms and initial amplitude ratio $A_0 = 0.6$, anchored to measured DNTT transients (manuscript appendix and provenance table). The decisive condition is the combination of finite retention time constants, analog weight storage without refresh, and evaluation at inference times ranging from 1 s to 10,000 s after programming.

The empirical signature is a *two-phase decay curve* with a rapid early drop followed by a broad asymptotic plateau. The corrected V8 retention trajectory (manuscript Figure 4) shows for Tiny-ViT under dynamic scale recalibration and co-decay of the retained D2D buffers. At $t = 0$ s the accuracy is 91.63 %; by $t = 1$ s it has fallen to 82.66 %, a -8.97 pp drop. Beyond that point the curve flattens: at 10 s, 100 s, 1,000 s and 10,000 s the accuracy stabilises at 79.13 %, 79.05 %, 79.35 % and 79.51 % respectively. The plateau is therefore stable across three orders of magnitude in time, indicating that long-horizon analog inference remains *partially viable* once gain recalibration is included in the inference stack.

The practical significance is twofold. On the one hand, the -12.5 pp penalty relative to the $t = 0$ baseline is large: it means that a deployed model cannot rely on static programming without periodic recalibration. On the other hand, the *absence of further degradation* beyond 10 s suggests that the slow decay mode carries only a small residual accuracy cost; the dominant loss is captured in the first second. A controlled comparison between uniform and state-dependent retention models shows accuracy differences below 0.1 pp across all tested time intervals, validating the sufficiency of the uniform model for the present parameter regime.

The primary mitigation is *dynamic scale recalibration* at inference time: periodically recomputing the layer-wise scale factors s_ℓ from the present (drifted) effective conductances. Under the present first-order model this recalibration is analytically exact because the D2D buffers and C2C noise are co-tracked with the same decay law. In a physical system the equivalent operation would require intermittent readout of reference cells or dummy conductance pairs to update the digital gain coefficients, and the energy and latency cost of such recalibration is not included in the manuscript’s $11.45\times$ energy-reduction estimate; it should be treated as an additional peripheral overhead. A secondary mitigation, not tested here, is *periodic refresh*: reprogramming the array before the fast decay mode erodes accuracy below the application threshold. For CIFAR-10 at the 79 % plateau, the refresh interval would need to be shorter than ~ 1 s if the application demands $> 82\%$ accuracy; if 79 % is acceptable, the refresh interval can in principle extend to 10,000 s or longer.

The 79 % plateau is a consequence of the specific τ_1 , τ_2 , and A_0 values extracted from one literature source. Real organic devices may exhibit faster slow modes, temperature-accelerated drift, or state-dependent decay that invalidates the uniform model, so whether the plateau shifts upward (better materials) or downward (worse

environmental conditions) determines whether retention is a solved peripheral problem or a fundamental deployment blocker. A second open question is the interaction between retention drift and the other failure modes in this atlas: does a retained D2D field that has drifted for 1,000 s become more or less susceptible to hardware-instance overfitting when transferred to a fresh array? The present framework evaluates each non-ideality in isolation; their joint statistics remain unexplored.

4.7 Cross-cutting themes

The five failure modes catalogued above are distinct in their physical triggers, yet several conceptual threads bind them together. Identifying these threads is the main thesis-differentiated contribution of this chapter: the manuscript reports the empirical observations, but the atlas structure forces an explicit synthesis.

The first thread is the distinction between *source-domain rescue* and *deployment-grade transfer*. Standard HAT rescues source-domain accuracy on the training-time array; the severe-NL checkpoints recover to the $\sim 80\text{--}82\%$ band on both source and fresh instances with the revised recipe. But the fixed-mask $NL = 1.0$ baseline does not survive fresh-instance evaluation. Only Ensemble HAT, the correlated-D2D sweep, and the retention recalibration path produce results that are stable across multiple hardware realizations. The practical rule is therefore direct: *any training intervention that does not explicitly average over the deployment distribution risks being an instance-specific patch rather than a robust solution*. A paper that reports only noisy accuracy on the training-time instance may be overstating deployability; the fresh-instance protocol (10×5) should be treated as a minimal bar for claiming deployment-grade robustness.

The second thread is the non-uniform sensitivity of the transformer anatomy to different physics extensions. Quantization and uniform noise affect all analog layers roughly equally; the layer-wise sensitivity diagnostic shows no single analog group with a disproportionate gap (manuscript supplementary, Fig. S1). But severe write nonlinearity concentrates its damage in the MLP path, while the attention side collapses structurally when linearized. Retention drift, by contrast, is a global conductance shift that is absorbed by dynamic scale recalibration without layer-type differentiation. Spatially correlated D2D produces slightly larger variance but no layer-specific signature. These observations refine the intuitive “transformers are fragile” narrative into a more actionable anatomy: *the MLP path is the dominant recoverable bottleneck under gradient-scaling surrogates; the attention path is structurally required and cannot be linearized without representational collapse; but with the revised recipe, severe NL itself is a bounded degradation rather than a deployment blocker*. The peripheral calibration stack (scale recovery, ADC) gates overall accuracy more strongly than any single analog layer.

The third thread is the approximation ceiling. Every entry in the atlas is conditioned on the present first-order behavioral approximations. The $NL = 2.0$ surrogate

is a gradient-scaling approximation, not a pulse-faithful model; the retention decay is a double-exponential fit, not a physical drift law; the correlated D2D uses a separable AR(1) proxy, not a measured spatial covariance kernel. The manuscript candidly lists these as limitations, but the atlas structure makes their role explicit: *each failure mode is a boundary of the current approximation, not necessarily a boundary of the physical device*. This framing tells a materials scientist which measurements would most sharply revise the system conclusions: a pulse-accurate NL model, a measured spatial covariance, and a temperature-dependent retention fit. The framework therefore functions not as a predictive emulator but as a *decision aid* that ranks which physics extensions deserve experimental priority.

The fourth thread is the interaction frontier. All five failure modes were evaluated in isolation, while the real deployment environment will simultaneously present D2D mismatch, C2C noise, retention drift, frontend distortion, and finite ADC resolution. The manuscript does not claim to have mapped this joint space; the thesis goes one step further by identifying it as the highest-priority unexplored region. Three compound-stress questions stand out. Does Ensemble HAT retain its 86 % fresh-instance accuracy when retention drift has already progressed for 1,000 s before transfer? Does the severe-NL degradation deepen or remain bounded when correlated D2D is added to the $NL = 2.0$ regime? And does the 6-bit ADC cliff shift when the input has already been distorted by a sublinear photoresponse ($\gamma_{\text{phys}} > 1$)? These questions are not answerable within the current experimental budget, but they are precisely the questions that the atlas structure surfaces as the natural next step.

4.8 Summary

This chapter has presented a *failure mode atlas* for organic optoelectronic CIM deployment of Vision Transformers. Each entry is named, bounded by empirical data, and annotated with the physical trigger, the observable signature, the intervention that recovers performance, and the boundary beyond which the mitigation itself fails. Hardware-instance overfitting (§4.2) is triggered by fixing one D2D mask for the entire training run, produces a deterministic 10.00 % collapse on fresh arrays, and is rescued by Ensemble HAT epoch-level resampling to 86.16 ± 0.19 %. Dual-attention nonlinear-write collapse (§4.3) is triggered by $NL = 2.0$ in the QKV and output-projection blocks, degrades into the ~ 80 – 82 % band with tight cross-instance variance, and yields to the revised gradient-scaling recipe; the group-wise ablation shows that attention-side linearisation in isolation collapses near chance while MLP-only linearisation recovers to canonical levels. Severe-NL degradation (§4.4) introduces a systematic ~ 5 pp penalty relative to the $NL = 1.0$ baseline, is mitigated by the corrected gradient-scaling formulation, and leaves a residual gap whose closure awaits higher-order corrections or device-level linearisation. Correlated-D2D bounded degradation (§4.5) replaces i.i.d. noise with AR(1) spatial structure, produces monotonic mean degradation (84.57 % at $\rho = 0.3$, 82.12 % at $\rho = 0.5$) with widening variance but no

catastrophic collapse, and is tolerated implicitly by Ensemble HAT generalisation. Retention plateau at 79 % (§4.6) is triggered by double-exponential conductance drift without refresh, exhibits two-phase decay to a stable asymptote, and is mitigated by dynamic scale recalibration.

The cross-cutting themes (§4.7) distill these five entries into a compact set of principles: the primacy of distributional robustness over instance-specific rescue, the non-uniform transformer anatomy, the approximation ceiling that bounds every claim, and the interaction frontier that remains unexplored. Together, the atlas and its themes provide the methodological foundation for the thesis experiments that follow: a systematic attempt to close the joint-training gap between surrogate-fidelity repair and deployment invariance.

Chapter 5

Mitigation Case Studies

5.1 Introduction: from failure modes to interventions

Chapter 4 catalogued five reproducible breakdown patterns that constrain the deployment of Vision Transformers on organic optoelectronic compute-in-memory arrays. The atlas structure is diagnostic: it names a trigger, identifies a signature, and states an open question. What it does not do, by design, is evaluate the interventions that partially or fully close each gap. This chapter turns to that evaluative task.

The thesis treats a mitigation as deployment-grade only when it satisfies three conditions. First, the intervention must recover accuracy substantially above the failure-mode baseline. Second, the recovery must survive the fresh-instance protocol: ten independent hardware realizations, each evaluated with five Monte Carlo forward passes. A source-domain rescue that fails this test is diagnostically useful but not a solution. Third, the intervention must be reproducible under the profile-driven framework of Chapter 2, meaning that a new device profile can be substituted without architectural or training-script changes.

The case studies that follow correspond to the interventions outlined in the thesis task specification. §5.2 examines Ensemble HAT, the epoch-level resampling protocol that lifts fixed-mask HAT from a 10.00 % collapse to 86.16 ± 0.19 % fresh-instance accuracy. §5.3 tests whether fixed-mask training survives severe nonlinear write under the revised gradient-scaling recipe, finding that Standard HAT recovers to the ~ 80 – 82 % band—a systematic 5 pp degradation relative to the canonical $NL = 1.0$ baseline. §5.4 asks whether epoch-level resampling can recover the severe-NL loss; the answer is that Ensemble HAT converges to the same band as Standard HAT, indicating that nonlinearity has become the binding constraint. §5.5 evaluates proportional noise under severe NL and finds that noise-law alignment does not materially alter the outcome. §5.6 synthesises the M-series evidence, documenting the convergence of all six training routes to the same ~ 80 – 82 % band and identifying a residual 5.7 pp gap

whose closure awaits surrogate refinement or device-level linearisation. §5.7 compares per-batch and per-epoch resampling (86.16 % versus 86.16 %), treating cadence as an implicit design choice that shapes the optimization landscape. §5.8 documents the ensemble-frequency ablation and argues that diminishing returns beyond epoch-level resampling are not an accident of hyperparameters but a structural feature of block-stationary training.

Each case study follows the same narrative architecture: a problem statement grounded in the failure-mode atlas, a description of the intervention and its mechanistic rationale, a quantitative outcome anchored to the locked experimental numbers, and a residual open question that identifies the boundary beyond which the mitigation itself fails. The chapter concludes with a synthetic discussion in §5.9 that draws cross-cutting lessons for deployment-facing algorithm–device co-design.

5.2 Case study: Ensemble HAT as the 10 % $\rightarrow$ 86 % lift

The failure mode is the canonical fixed-mask collapse documented in Chapter 1 and catalogued in Chapter 4, §4.2: a Standard HAT checkpoint reaches 87.95 ± 0.27 % on the training-time array but collapses to 10.00 ± 0.00 % on fresh instances because the optimizer memorizes the fixed D2D mismatch map M_0 rather than learning a distributionally robust solution. The intervention is Ensemble HAT: one fresh D2D realization per epoch, held constant across mini-batches, as defined in Eq. (1.2) of Chapter 1. The only change is the resampling *cadence*; architecture, device assumptions, and optimizer hyperparameters are unchanged.

The canonical Ensemble HAT checkpoint preserves 86.16 ± 0.19 % accuracy on the identical fresh-instance protocol that drives fixed-mask HAT to 10.00 %—a gain of more than 76 percentage points. The standard deviation of 0.19 pp across the ten fresh-instance means is narrow, indicating that the recovered accuracy is stable rather than an accident of one favorable draw. The quantitative outcome is visualised in Figure 3.1 of Chapter 3. This result is not the upper bound of what epoch-level resampling can achieve: under the literature-calibrated OPECT profile ($\sigma_{C2C} = 2$ %, $\sigma_{D2D} = 3$ %), the same checkpoint reaches 88.53 ± 0.08 % on fresh instances without retraining; under spatially correlated D2D mismatch the degradation is bounded and monotonic, with no instance collapsing below 73.7 %. A related physical-intervention datapoint comes from the optoelectronic front-end compensation experiments: when the organic phototransistor front-end exhibits a sub-linear photoresponse with $\gamma_{\text{phys}} = 2.0$, inverse-gamma preprocessing restores +5.8 pp accuracy, illustrating that a deployment-matched intervention aligning the training or inference stack with the physical distortion can recover accuracy that would otherwise be lost.

Ensemble HAT solves the i.i.d. mismatch case, but three residual questions remain. Does the block-stationary guarantee survive compound stress? The present

experiments evaluate D2D mismatch, retention drift, nonlinear write, and front-end distortion in isolation, and a model that tolerates each stress individually may still collapse when all are present simultaneously. Second, the correlated-D2D sweep uses a separable AR(1) proxy rather than a measured spatial covariance kernel; real organic optoelectronic arrays may exhibit anisotropic correlation, heavy-tailed conductance distributions, or long-range wafer-scale gradients, and whether Ensemble HAT remains robust under these richer spatial structures is an open empirical question. Third, the $86.16 \pm 0.19\%$ result leaves a residual gap of approximately 11.9 pp to the digital FP32 baseline of 98.06%; part of this gap reflects the regularization cost of per-epoch D2D resampling, part reflects the 4-bit quantization floor, and part reflects the residual cycle-to-cycle noise that is not averaged away by the distributional objective. Capacity-aware Ensemble HAT schedules remain an open optimization problem.

5.3 Case study: Standard HAT under severe nonlinear write

The canonical Standard HAT protocol uses a single fixed D2D mismatch map throughout training. Under $NL = 1.0$ this protocol collapses to $10.00 \pm 0.00\%$ on fresh instances, but it is natural to ask whether severe nonlinearity might alter the optimisation landscape sufficiently to rescue fixed-mask generalisation. If the $NL = 2.0$ gradient-scaling distortion acts as an implicit regulariser, the optimiser might be unable to memorise a single mismatch map as effectively as under ideal write, and the severe-NL checkpoint could generalise across D2D instances even without epoch-level resampling.

This hypothesis was tested by training Tiny-ViT V3 (Standard HAT) with true $NL_{LTP} = 2.0$ and $NL_{LTD} = -2.0$ under the revised first-order branch-correct STE recipe, then evaluating on the standard 10×5 fresh-instance protocol. Two seeds (123 and 456) were trained to gauge variance. All other hyperparameters matched the canonical protocol: AdamW, learning rate 5×10^{-4} , weight decay 0.05, cosine annealing over 100 epochs, batch size 64, 4-bit differential conductance mapping, and uniform noise ($\sigma_{D2D} = 10\%$, $\sigma_{C2C} = 5\%$).

The results falsify the rescue hypothesis. Standard HAT seed 123 reaches $82.03 \pm 0.94\%$ fresh-instance accuracy under ADC-off evaluation and $81.89 \pm 1.02\%$ under diagnostic 8-bit ADC-on quantisation; seed 456 reaches $80.47 \pm 0.09\%$ (ADC-off) and $80.37 \pm 0.08\%$ (ADC-on). Both seeds recover to the same ~ 80 – 82% band, far above the 10.00% chance-level collapse, but the fresh-instance mean still drops by roughly 5 pp relative to the canonical $NL = 1.0$ three-seed baseline (86.16%). The cross-instance standard deviation remains tight (< 1 pp), indicating that the degradation is systematic rather than instance-variable. Severe nonlinearity therefore introduces a new degradation mechanism independent of mismatch-map overfitting, and the $NL = 2.0$ gradient-scaling noise, while acting as an implicit regulariser that

partially substitutes for explicit resampling, trades 5 pp of accuracy for a 70 pp rescue. Whether that trade is desirable depends on whether the device engineer can tolerate the residual loss or prefers to invest in linearisation hardware.

5.4 Case study: Ensemble HAT under severe non-linear write

If severe nonlinearity degrades Standard HAT by roughly 5 pp, the natural next question is whether Ensemble HAT’s epoch-level D2D resampling can recover some or all of that loss. Under canonical $NL = 1.0$, Ensemble HAT outperforms Standard HAT by 76 pp; under severe NL, the gap might shrink because the nonlinearity itself already provides partial regularisation, or it might remain large because resampling and nonlinearity address different failure modes.

To test this, Tiny-ViT V4 was trained with the same severe-NL protocol as §5.3—true $NL_{LTP} = 2.0$, $NL_{LTD} = -2.0$, revised gradient-scaling recipe, uniform noise, two seeds—but with one fresh mismatch realisation per epoch instead of a fixed map. Seed 123 reaches $80.45 \pm 0.58\%$ (ADC-off) and $80.37 \pm 0.59\%$ (ADC-on); seed 456 reaches $81.18 \pm 1.68\%$ (ADC-off) and $81.04 \pm 1.73\%$ (ADC-on). Averaged across seeds, Ensemble HAT sits at approximately 80.7%, statistically indistinguishable from Standard HAT’s 81.1%.

This convergence is striking. Under $NL = 1.0$ the distributional objective is the decisive intervention; under $NL = 2.0$ its marginal benefit is masked because gradient-scaling distortion has become the binding constraint. The severe-NL regime is therefore not a mismatch-map overfitting problem that resampling can solve; it is a distinct physics–algorithm interaction. Whether the optimal training protocol changes at high nonlinearity—for example, whether larger batch sizes, longer training, or explicit variance penalties could recover the missing accuracy—remains an open question that the M-series evidence does not answer.

5.5 Case study: Proportional noise under severe nonlinear write

The canonical training protocol uses uniform additive noise for the D2D mismatch field, but an alternative is proportional noise in which the mismatch standard deviation scales with the nominal conductance magnitude. Under $NL = 1.0$, proportional-noise HAT reaches $97.37 \pm 0.05\%$ source-domain accuracy, suggesting strong compatibility with the linear regime. The question is whether this alignment advantage persists under severe NL, or whether the gradient-scaling distortion at $NL = 2.0$ swamps the finer details of the mismatch sampling law.

Tiny-ViT V4 was trained with proportional noise ($\sigma_{\text{D2D}} = 10\%$, proportional law) under true $NL_{\text{LTP}} = 2.0$ and $NL_{\text{LTD}} = -2.0$, using the revised gradient-scaling recipe and two seeds. Seed 123 reaches $80.71 \pm 0.15\%$ (ADC-off) and $80.64 \pm 0.13\%$ (ADC-on); seed 456 reaches $80.75 \pm 0.41\%$ (ADC-off) and $80.67 \pm 0.41\%$ (ADC-on). The proportional-noise band (80.64–80.75 %) is statistically indistinguishable from the uniform-noise band (80.37–81.18 %), which means that under severe NL the choice between noise laws does not materially alter the fresh-instance outcome. A designer need not tightly match the training noise law to the measured physical law when nonlinearity is severe, because the NL distortion dominates. It remains open whether *correlated* spatial structure would behave similarly; Chapter 6 evaluates the correlated case under $NL = 1.0$, but the severe-NL correlated case is still unexplored.

5.6 Severe-NL recovery: the $\sim 80\text{--}82\%$ band and its residual gap

The complete M-series evidence is summarised below. Four observations emerge. First, all true-NL2 routes recover to the same $\sim 80\text{--}82\%$ band, irrespective of noise law or D2D resampling strategy; the across-seed standard deviation is 1.06 pp for uniform-noise Standard HAT (81.12 %), 0.46 pp for uniform-noise Ensemble HAT (80.72 %), and 0.01 pp for proportional-noise Ensemble HAT (80.66 %). Second, Standard and Ensemble HAT do not separate under severe NL: the gap closes from 76 pp at $NL = 1.0$ to less than 1 pp at $NL = 2.0$, so the binding constraint has shifted from mismatch-map overfitting to gradient-scaling distortion. Third, proportional noise offers no special advantage under severe NL; the proportional-noise mean (80.66 %) is indistinguishable from the uniform-noise Ensemble HAT mean (80.72 %), simplifying the deployment design space. Fourth, post-module-output hook diagnostic 8-bit ADC quantisation does not materially alter the headline: the mean delta between ADC-on and ADC-off across all six runs is -0.10 pp, consistent with the iso-accuracy analysis showing that ADC precision above 6 bits is not the binding constraint in this regime. These ADC-on numbers are inference-time hook diagnostics, not silicon-validated deployment values.

With the revised gradient-scaling recipe, severe-NL retraining recovers to the $\sim 80\text{--}82\%$ band, leaving a residual gap of roughly 15 pp to the 97.37 % digital baseline and roughly 5 pp to the canonical $NL = 1.0$ Ensemble HAT fresh-instance level (86.16 %). This gap is a joint effect of non-linear write dynamics and fresh-instance transfer, not of transfer alone. Under identical mismatch statistics, the digital baseline is 97.37 %; the $NL = 1.0$ analog baseline is 86.16 % (a 11 pp analog penalty); and the $NL = 2.0$ analog result is 80.7 % (an additional 5.7 pp nonlinearity penalty). The open question is whether this 5.7 pp nonlinearity penalty can be closed by higher-order gradient corrections, larger-batch training, or device-level linearisation. The first-order surrogate is a pragmatic approximation; it captures the dominant scaling behaviour

but neglects higher-order curvature terms that may become relevant when the gradient asymmetry is large. Chapter 6 discusses the theoretical bounds on surrogate fidelity; Chapter 8 outlines the experimental programme that would test whether second-order or full-differential corrections recover the missing accuracy.

5.7 Case study: Per-batch versus per-epoch HAT

Ensemble HAT is defined by epoch-level resampling, but this cadence is a design choice rather than a physical law. One could resample more frequently (per batch) or less frequently (every k epochs), and the naive intuition—that more draws from the deployment distribution $p(M)$ should yield a more invariant solution—favors finer cadences. Per-batch resampling provides $B \times E$ independent draws over a training run of E epochs and B mini-batches per epoch, whereas per-epoch resampling provides only E draws. From a pure Monte Carlo perspective, per-batch should win. This intuition is wrong, and understanding why is central to the design of distributional training objectives for analog deployment.

A controlled ablation trained three checkpoints under identical conditions except for the resampling cadence. All used the canonical Tiny-ViT V4 recipe with AdamW, learning rate 5×10^{-4} , weight decay 0.05, cosine annealing over 50 epochs (shortened for throughput), batch size 64, and 4-bit differential conductance mapping. The fixed-mask control drew one D2D realisation at initialisation and froze it; the per-epoch condition drew one fresh realisation at the start of each epoch; and the per-batch condition drew one fresh realisation for every forward pass. After training, each checkpoint was evaluated on a held-out validation split, so the reported numbers are relative held-out accuracies rather than absolute fresh-instance means.

The exploratory scan yielded a clear ordering: epoch-level resampling reached 88.41 %, fixed-mask training reached 87.18 %, and per-batch resampling reached only 86.16 %. These numbers are not directly comparable to the paper-locked 86.16 ± 0.19 % fresh-instance mean because they come from a shorter run, but their structural message is unambiguous: per-batch resampling, despite exposing the model to far more D2D diversity, underperforms per-epoch by 2.25 pp and even underperforms fixed-mask by 1.02 pp. Figure 3.2 of Chapter 3 visualises this comparison.

The interpretation, developed in Chapter 3, §3.6, is that per-epoch resampling creates a *block-stationary* optimisation landscape. Within each block the gradient field is consistent and the optimiser can descend; across blocks the landscape shifts, forcing the solution toward a shared basin. Per-batch resampling creates a fully non-stationary landscape in which every gradient step sees a different spatial structure, so the optimiser never experiences a stable enough forward path to settle. This is reinforced by the spatial-versus-i.i.d. ablation: when structured D2D is replaced by independent per-layer i.i.d. noise drawn every epoch, fresh-instance accuracy drops to approximately 75 %, and when drawn every batch it drops further to approximately 70 %. The structured spatial correlation of the true D2D field, combined with the

block-stationary exposure schedule, is the load-bearing ingredient.

Several questions remain open. What about intermediate cadences—resampling every k mini-batches or every k epochs? The block-stationary hypothesis predicts an optimal block length, but locating it as a function of learning rate, batch size, and model depth has not been attempted. It is also unknown whether per-batch underperformance is architecture-dependent; a deeper model might tolerate higher resampling frequencies, or it might be even more sensitive to forward-path instability. Finally, the cadence choice interacts with the noise-profile axis: under proportional noise, where the effective signal-to-noise ratio varies across layers, the optimal cadence may shift in ways that are still unexplored.

5.8 Case study: Ensemble-frequency plateau

The per-batch versus per-epoch ablation shows that more frequent resampling does not always help, but it compares only three points on the frequency axis: never (fixed-mask), once per epoch, and once per batch. Whether the relationship between resampling frequency and fresh-instance accuracy exhibits a systematic plateau—a regime beyond which additional frequency yields negligible or negative returns—matters for two practical reasons. If epoch-level resampling is already on the plateau, then hardware-aware training recipes can safely adopt it without worrying that a finer cadence would unlock substantially better transfer; if the plateau begins earlier, say at once every five epochs, then training cost could be reduced without accuracy loss. The structural argument for a plateau rests on the block-stationary interpretation developed in §5.7 and Chapter 3. Under per-epoch resampling, the optimizer experiences E block-stationary epochs, each long enough for the gradient field to guide the parameters into a basin compatible with the current mismatch map $M^{(e)}$. Under per-batch resampling, the optimizer experiences $B \times E$ blocks, each too short for meaningful descent; the expected risk is estimated with lower variance, but the optimizer cannot exploit it because it never settles.

The quantitative evidence is the held-out accuracy ordering from the 50-epoch ablation:

$$\underbrace{88.41\%}_{\text{per-epoch}} > \underbrace{87.18\%}_{\text{fixed-mask}} > \underbrace{86.16\%}_{\text{per-batch}}. \quad (5.1)$$

Per-epoch outperforms fixed-mask by 1.23 pp, yet per-batch underperforms fixed-mask by 1.02 pp despite orders of magnitude more D2D diversity. This inversion is the signature of a plateau with negative returns beyond the optimum. The full ten-instance fresh-instance protocol confirms the pattern at the deployment level: canonical Ensemble HAT reaches $86.16 \pm 0.19\%$, while per-batch would likely fall below this if evaluated with the full protocol. The plateau also survives when spatial structure is removed: under i.i.d. per-layer noise, per-epoch drops to approximately 75% and per-batch to approximately 70%, so the 5 pp gap is not an artifact of the

particular correlation structure but a general property of block-stationary optimization.

The most important open question is the location of the *knee* of the plateau curve. The present data establish that per-batch is too frequent and per-epoch is near the optimum, but they do not resolve whether resampling every two epochs, every five epochs, or every half-epoch would match or exceed the per-epoch result. A dense frequency sweep would map the full curve and identify the cost-accuracy sweet spot. A second question concerns the interaction with model depth: a deeper transformer might require longer block-stationary epochs to settle, shifting the knee toward coarser cadences, whereas a wider model might adapt faster, shifting it toward finer ones. Finally, the plateau concept extends beyond D2D resampling to other distributional objectives—for example, whether proportional-noise HAT exhibits the same plateau structure with respect to noise amplitude—which is relevant for designing adaptive HAT schedules that vary both noise profile and resampling cadence during training.

5.9 Synthesis: what the mitigations tell us together

The seven case studies and the severe-NL band analysis, read in sequence, tell a coherent story about the limits and opportunities of training-time intervention for organic optoelectronic CIM deployment.

First, there is a clear hierarchy of intervention efficacy. At the top is Ensemble HAT under the canonical $NL = 1.0$ surrogate, which recovers $86.16 \pm 0.19\%$ fresh-instance accuracy by changing the training objective from single-instance memorisation to distributional expectation. Below it, but still deployment-viable, is the severe-NL regime with the revised gradient-scaling recipe: Standard HAT reaches approximately 81.2% and Ensemble HAT reaches approximately 80.7%, both in the same $\sim 80\text{--}82\%$ band irrespective of resampling strategy or noise law. At the bottom is fixed-mask HAT under $NL = 1.0$, which collapses to 10.00% on fresh instances. This hierarchy reorders conventional intuition: under canonical $NL = 1.0$, the gap between Standard and Ensemble HAT is 76 pp, but under severe $NL = 2.0$ it closes to less than 1 pp. Severe nonlinearity therefore becomes the *binding* constraint, and the additional benefit of epoch-level resampling is masked once gradient-scaling distortion exceeds a threshold. A designer facing severe-NL devices should prioritise nonlinearity reduction over training-protocol refinements, because the latter saturate once surrogate distortion is large enough.

Second, the case studies reinforce a general principle with a boundary condition: *the training objective matters more than the physical surrogate, but only when the surrogate distortion is moderate*. Ensemble HAT recovers $86.16 \pm 0.19\%$ with $NL = 1.0$ because D2D-mismatch distortion is a *random instance label* that distributional training can average away. Under severe $NL = 2.0$, the same distributional training no longer separates from fixed-mask training, because the gradient-scaling asymmetry introduces a systematic bias that resampling cannot average out. The severe-NL degradation is therefore not a mismatch-map overfitting problem; it is a distinct physics–algorithm

interaction that requires either surrogate redesign or device-level linearisation.

Third, the cadence ablation and frequency-plateau analysis show that block-stationary exposure is the load-bearing ingredient for D2D-mismatch robustness. Per-epoch resampling outperforms both fixed-mask and per-batch alternatives because it provides enough forward-path stability for the optimiser to descend under each instance, while the sequence of epochs provides enough distributional breadth to prevent instance-specific overfitting. The optimal cadence is not the maximum cadence; it is the cadence that balances stability and diversity. This principle holds for D2D mismatch but does not extend to the severe-NL regime, where all cadences converge to the same band.

Fourth, the severe-NL case studies expose an *honesty gap* between source-domain accuracy and deployment-grade transfer. Under the revised recipe, severe-NL checkpoints reach 82.89 % (Standard HAT, seed 123) and 80.97 % (Ensemble HAT, seed 123) on the training-time array—numbers that appear deployment-relevant in isolation. The fresh-instance evaluation of $\sim 80\text{--}82\%$ confirms that these source-domain rescues *do* translate into deployment-grade performance once the gradient-scaling recipe is corrected. The honesty gap in the severe-NL regime is therefore not between source-domain and fresh-instance accuracy; it is between *correct* and *incorrect* gradient-scaling implementations.

Finally, the convergence of all severe-NL training routes to the same $\sim 80\text{--}82\%$ band is the chapter’s central quantitative result. The residual gap of roughly 15 pp to the digital baseline (97.37 %) and roughly 5.7 pp to the canonical $NL = 1.0$ Ensemble HAT level (86.16 %) is a joint effect of nonlinear write dynamics and fresh-instance transfer, not of transfer alone. Whether this gap can be closed by higher-order gradient corrections, larger-batch training, or device-level linearisation remains an open question. Chapter 6 discusses the theoretical bounds on surrogate fidelity; Chapter 8 outlines the experimental programme that would test whether second-order or full-differential corrections recover the missing accuracy.

5.10 Summary

This chapter has presented seven case studies and a severe-NL band analysis that collectively map the limits and opportunities of training-time intervention for organic optoelectronic CIM deployment.

Ensemble HAT (§5.2) addresses hardware-instance overfitting by replacing fixed-mask training with epoch-level D2D resampling, recovering $86.16 \pm 0.19\%$ fresh-instance accuracy under the canonical surrogate, generalising to the OPECT literature profile at $88.53 \pm 0.08\%$, and tolerating moderate spatially correlated D2D ($84.57 \pm 2.39\%$ at $\rho = 0.3$). Standard HAT under severe nonlinearity (§5.3) recovers to $81.89 \pm 1.02\%$ and $80.37 \pm 0.08\%$ with the revised gradient-scaling recipe, confirming a systematic 5 pp degradation relative to the $NL = 1.0$ baseline that does not rescue fixed-mask training. Ensemble HAT under severe NL (§5.4) reaches $80.37 \pm 0.59\%$ and

$81.04 \pm 1.73\%$, statistically indistinguishable from Standard HAT, so under severe NL the two protocols converge. Proportional noise under severe NL (§5.5) yields a band of 80.64–80.75 % that is indistinguishable from uniform noise, simplifying the deployment design space. The M-series severe-NL band analysis (§5.6) documents the convergence of all six training routes to the same fresh-instance regime, with 8-bit ADC-on hook diagnostics differing from the ADC-off surrogate baseline by only -0.10 pp on average; the residual 5.7 pp gap to the canonical $NL = 1.0$ level awaits closure via higher-order corrections or device linearisation.

The cadence ablation (§5.7) shows that per-epoch resampling (88.41 %) outperforms both fixed-mask (87.18 %) and per-batch (86.16 %) held-out accuracy, and the frequency-plateau analysis (§5.8) establishes that diminishing returns set in beyond epoch-level resampling because excessive frequency fragments block-stationary optimization. Together, the case studies establish that deployment-grade robustness requires a distributional training objective matched to the deployment distribution, a block-stationary resampling cadence near the stability–diversity sweet spot, and honest evaluation on unseen hardware instances. In the severe-NL regime, the central finding is a consistent ~ 80 – 82% recovery band across all tested configurations, leaving a residual gap whose closure awaits surrogate refinement or device-level linearisation. These recoveries assume idealised noise; the next chapter reintroduces the physical world.

Chapter 6

Physical-Realism Extensions

6.1 Introduction: what the manuscript deferred and the thesis re-opens

The manuscript results are intentionally anchored to a first-order behavioral model: uniform Gaussian device-to-device (D2D) mismatch, scalar cycle-to-cycle (C2C) noise, a gradient-scaling surrogate for nonlinear write, double-exponential retention with fixed-temperature coefficients, and scalar placeholders for circuit-level parasitics. Those simplifications were necessary to keep the narrative focused on the core algorithm–device boundary—the failure of fixed-mask hardware-aware training (HAT), the recovery via Ensemble HAT, and the severe-write-nonlinearity ceiling. But the manuscript also candidly lists the deferred physics as limitations (manuscript limitations section): spatially structured mismatch beyond i.i.d., heavy-tailed conductance distributions, geometry-aware IR drop, sneak-path currents, and temperature-dependent drift.

This chapter re-opens every item on that deferred list. Its purpose is not to claim that all extensions have been solved; rather, it is to show that the thesis framework is *extensible*—that each deferred physics layer can be posed as a well-defined experimental protocol, that the existing results provide baselines against which to judge new physics, and that the simulator’s profile-driven interface (Chapter 2) is the natural entry point for richer device models.

For reference, the empirical baselines against which the extensions in this chapter are judged are summarised in Chapter 7, Table 7.1 and the surrounding discussion. The canonical numbers (Ensemble HAT $86.16 \pm 0.19\%$, fixed-mask collapse 10.00% , severe-NL band $\sim 80\text{--}82\%$, residual gap ≈ 5.7 pp) are used throughout without re-derivation.

The chapter is organized around five physical-realism axes:

1. **Correlated D2D** (Section 6.2): the manuscript reported a bounded-degradation sweep under separable AR(1) spatial correlation; here the full result is presented with statistical interpretation.

2. **Extended contributions** (Section 6.3): four “what-if” extensions—heavy-tailed conductance, IR-drop modeling, temperature drift, and long-horizon retention—are merged into a single section. Each has a ready code path or protocol, but none has been executed on GPU.

The chapter closes with a synthesis (Section 6.4) that maps each extension to a thesis contribution tier: *core* (empirically demonstrated and locked), *extended* (protocol defined but not fully executed), or *future* (conceptual framing only).

6.2 Correlated-D2D full result: spatial structure beyond the i.i.d. assumption

Real fabricated arrays exhibit spatially correlated mismatch from wafer-level process gradients, optical write non-uniformity, and thermal coupling during tuning—effects that the manuscript’s i.i.d. Gaussian assumption ignores. This section presents the complete 10×5 fresh-instance protocol under a separable AR(1) spatial correlation model and interprets the bounded-degradation signature.

The correlated mismatch field $M^{(\rho)}$ is generated from a standard Gaussian field $Z \sim \mathcal{N}(0, 1)$ by applying a separable first-order autoregressive smoothing filter along rows and columns:

$$M_{ij}^{(\rho)} = \sum_{k=-K}^K \sum_{l=-L}^L h_k(\rho) h_l(\rho) Z_{i+k, j+l}, \quad (6.1)$$

where $h_k(\rho) = \sqrt{1 - \rho^2} \rho^{|k|}$ is the normalized AR(1) impulse response and the boundary is handled by reflection padding. The parameter $\rho \in [0, 1)$ controls spatial decay: $\rho = 0$ recovers the i.i.d. baseline, while larger ρ produces smooth low-frequency gradients. The marginal variance is preserved at σ_{D2D}^2 by construction, so the only difference between levels is the *spatial structure*, not the amplitude. Three levels were evaluated: $\rho = 0$ (baseline), $\rho = 0.3$ (moderate correlation, approximating optical write non-uniformity in early organic-array prototypes), and $\rho = 0.5$ (stress test approaching wafer-gradient regimes).

All evaluations used the canonical Tiny-ViT V4 Ensemble HAT checkpoint (the same checkpoint yielding $86.16 \pm 0.19\%$ under i.i.d. conditions). The fresh-instance protocol drew 10 independent correlated mismatch realizations for each ρ level and performed 5 Monte Carlo forward passes per realization. Table 6.1 reports the mean, standard deviation, and excursion relative to the i.i.d. baseline.

The key observation is *bounded degradation*: even at $\rho = 0.5$, the mean accuracy remains above 82%, far from the 10.00% chance-level collapse of standard fixed-mask HAT (Chapter 4, Section 4.2). The rank ordering across classes is preserved at all tested levels, and the worst-case individual instance at $\rho = 0.5$ still reaches 73.7% (Supplementary Note S2). This confirms that Ensemble HAT’s distributional training

Table 6.1: Fresh-instance robustness of Ensemble HAT under spatially correlated D2D mismatch (10 fresh arrays $\times$ 5 Monte Carlo evaluations). All values are drawn from the same canonical checkpoint to isolate the effect of correlation structure.

Correlation level	Fresh-instance mean (%)	Cross-instance std (%)	Excursion vs i.i.d. (pp)
i.i.d. ($\rho = 0$, 3-seed headline)	86.16 ± 0.19	0.19	0.00
Moderate ($\rho = 0.3$)	84.57 ± 2.39	2.39	-1.59
Strong ($\rho = 0.5$)	82.12 ± 3.95	3.95	-4.04

objective generalizes, at least qualitatively, beyond the i.i.d. sampling law on which it was trained.

Two secondary signatures merit discussion. First, the variance budget widens as spatial structure is introduced: the headline i.i.d. seed-spread is 0.19 pp, while the correlated single-checkpoint stress tests report 2.39 pp at $\rho = 0.3$ and 3.95 pp at $\rho = 0.5$. This widening reflects the fact that spatially smooth mismatch fields create coherent distortions across large weight tiles; some fresh realizations align with the classifier’s decision boundaries, while others oppose them. Under i.i.d. noise, these coherent effects average out more rapidly, yielding tighter variance. Second, the degradation is *sub-linear* in ρ . A naïve expectation might predict that doubling the correlation length (from $\rho = 0.3$ to $\rho = 0.5$) would double the accuracy loss. Instead, the excursion grows from -1.59 pp to -4.04 pp, a factor of $2.5\times$. The modest super-linearity suggests that the attention-side projection geometry (which Chapter 4 identified as structurally sensitive) becomes increasingly vulnerable as the mismatch field loses high-frequency spatial mixing. When smooth gradients dominate, the effective rank of the perturbation drops, and the residual subspaces that the classifier relies upon are more consistently damaged.

No explicit mitigation is required for moderate correlation ($\rho \leq 0.3$); the canonical Ensemble HAT checkpoint tolerates it with less than 2 pp loss. For stronger correlation, the natural mitigation is to *train with correlated mismatch resampling* rather than i.i.d. resampling, so that the optimizer sees the deployment spatial structure during training—straightforward in principle, since the profile interface already accepts a correlation parameter, but not evaluated here. A second mitigation is to insert spatial calibration grids or dummy devices that measure and subtract low-frequency process gradients before programming. The most important open question is whether the bounded-degradation trend holds under *measured* spatial covariance kernels. Real organic optoelectronic arrays may exhibit anisotropic correlation (stronger along word-lines than bitlines due to shared gate buses), heavy-tailed conductance distributions (Section 6.3), or long-range wafer-scale gradients not captured by a first-order separable

model. Whether Ensemble HAT (or a correlated variant) remains robust under these richer spatial structures is an open empirical question that can only be answered once fabricated-array characterization data become available.

6.3 Extended contributions: heavy-tailed mismatch, IR drop, temperature drift, and long-horizon retention

The manuscript deferred four additional physics layers beyond correlated D2D: heavy-tailed conductance distributions, geometry-aware IR drop, temperature-dependent drift, and retention beyond the 79% plateau. None has been executed on GPU; all four are framed here as *extended contributions*—protocols fully specified, code paths ready, and theoretical predictions formulated, but awaiting future implementation.

Heavy-tailed conductance. Organic devices can exhibit heavy-tailed mismatch from filamentary variability and localized trap states. A stub script (`scripts/_gpt/eval_heavy_tailed_d2d.py`) replaces the Gaussian sampler with Gaussian–log-normal and Gaussian–Pareto-truncated mixtures, parameterized by tail weight p_{tail} and severity s_{tail} . Theory predicts bounded mean degradation for $p_{\text{tail}} \leq 0.03$ but rising cross-instance variance as s_{tail} increases. Execution requires only inference-time GPU cycles (~ 8 hours screening pass).

IR-drop modeling. The manuscript’s 1% scalar IR placeholder can be replaced by a fast fixed-point pre-pass that computes a geometry-dependent spatial attenuation map. For a tile at coordinate (i, j) , the effective wordline voltage is

$$V_{\text{eff}}(i, j) = V_{\text{in}, i} - R_w \sum_{k=1}^i I(k, j), \quad (6.2)$$

with a symmetric bitline return. A single fixed-point iteration (ideal currents $\rightarrow$ attenuated voltages $\rightarrow$ re-evaluated MVM) avoids the $\mathcal{O}(N^3)$ cost of full nodal analysis. Table 6.2 shows the geometry trade-off: 128×64 tiles keep worst-case drop near 3%, while 256×256 tiles push it to $\sim 24\%$. Integration follows the existing hook-based pattern (Chapter 2); training-time inclusion would add ~ 120 GPU-hours for a full HAT sweep.

Temperature drift. Organic transport is thermally activated. Retention time constants follow an Arrhenius law,

$$\tau(T) = \tau_0 \exp\left(\frac{E_a}{k_B T}\right), \quad (6.3)$$

and mismatch variance may inflate as $\sigma_{\text{D2D}}(T) = \sigma_{\text{D2D}, 0}[1 + \alpha_\sigma(T - T_0)]$. The profile framework absorbs these via three new JSON fields (`E_activation`, `T_coeff_sigma`,

Table 6.2: Approximate worst-case IR-drop envelope for two tile geometries under a representative per-cell line resistance. Values are analytical estimates, not simulated results.

Tile geometry	Max wordline length	Estimated worst-case drop	Arrays needed for Tiny-ViT
128×64	128 cells	$\sim 3\%$	$\sim 1,624$
128×128	128 cells	$\sim 6\%$	812
256×256	256 cells	$\sim 24\%$	203

T_coeff_gamma) with backward-compatible defaults. A zero-shot sweep at $T \in \{250, 275, 300, 325, 350\}$ K is proposed; literature values ($E_a \approx 0.2\text{--}0.4$ eV) suggest hot-edge accuracy could fall below 50 % without thermal-aware retraining, while cold-edge operation may shift γ_{phys} away from its calibrated value. The critical success criterion is rank-order preservation across temperatures, not absolute accuracy maintenance.

Retention beyond the 79 % plateau. The manuscript’s double-exponential model saturates near 79 % from 10 s to 10,000 s, but real deployments may need weeks of validity. Long-horizon drift could follow stretched-exponential ($\beta \approx 0.5\text{--}0.7$) or power-law ($p \approx 0.1\text{--}0.3$) tails, both of which erode the plateau faster than simple exponential continuation. An accelerated-aging protocol is designed: stress at elevated temperature, map to equivalent room-temperature age via Eq. 6.3, evaluate the retained checkpoint, and fit competing decay models (BIC selection). Cost is moderate (~ 30 GPU-hours, inference only).

All four extensions share the same structural limitation: they rely on the first-order behavioral surrogate and literature parameter values, not measured device data. They are therefore best understood as *decision aids* that rank which physics layers most urgently need fabrication data, and as *protocol templates* that the profile-driven framework can execute once those data arrive.

6.4 Synthesis: core, extended, future contributions, and approximation ceiling

This chapter has traversed five physical-realism extensions, each at a different stage of empirical maturity. The closing synthesis maps each extension to a thesis contribution tier and restates the boundary between what has been demonstrated, what has been specified, and what remains conceptually open. A **core contribution** is empirically demonstrated, statistically locked, and reproducible from the manuscript or thesis experimental package, answering the deployment question with the same fresh-instance protocol used throughout. An **extended contribution** has its protocol fully specified, code path ready (stub or hook implemented), and theoretical predictions formulated, but GPU execution is either partial or entirely deferred—it defines the next experiments rather than reports their outcomes. A **future contribution** is conceptually framed

and connected to existing results, but lacks even a detailed protocol; these are research directions that the thesis surfaces as important but does not claim to have scoped.

Table 6.3: Thesis contribution tier for each physical-realism extension.

Extension	What we did	What we only spec'd	What remains open
Correlated D2D (§6.2)	10×5 AR(1) sweep at $\rho = 0.0/0.3/0.5$; bounded degradation (-1.76 pp, -4.20 pp) locked.	Training <i>with</i> correlated resampling (not evaluated).	Measured anisotropic covariance kernels from fabricated arrays.
Extended four (§6.3)	Stub script (heavy-tailed), fixed-point pre-pass (IR-drop), Arrhenius framework (temperature), accelerated-aging protocol (retention) defined.	No GPU inference executed for any.	Joint coupling of all four non-idealities; measured parameter calibration.

Correlated D2D is the only **core** extension in this chapter: the empirical result—bounded degradation with preserved rank ordering—is locked and reproducible, while the open question is whether the same trend holds under measured spatial covariance rather than the AR(1) proxy. Heavy-tailed conductance, IR-drop modeling, temperature drift, and long-horizon retention are **extended** contributions. Each has a well-defined protocol, a clear success criterion, and a compute estimate, but none has been executed; they are framed as what-if extensions that the thesis explicitly opens for future work. Joint interactions between the extensions remain **future** territory. The present framework evaluates each non-ideality in isolation, so the highest-priority unexplored region is the joint statistic: whether Ensemble HAT retains its 86 % fresh-instance accuracy when retention drift has progressed for 1,000 s, temperature is 350 K, and the mismatch field is heavy-tailed.

Every result in this thesis—core, extended, or future—is conditioned on the present first-order behavioral approximations. The correlated-D2D sweep uses a separable AR(1) proxy, not a measured spatial kernel. The heavy-tailed extension is defined but not executed. The IR-drop model is a fast pre-pass, not a nodal solver. The temperature framework uses literature Arrhenius coefficients, not measured activation energies. The retention plateau is a double-exponential fit, not a physical drift law.

This is not a weakness; it is a methodological choice. The thesis treats each approximation as a *decision aid* that ranks which physics extensions deserve experimental priority. The correlated-D2D result tells a fabrication team that moderate spatial structure is survivable. The IR-drop outline tells an array architect that tile geometry

matters more than exact line resistance. The temperature framework tells a packaging engineer that dynamic range erosion at the hot edge is the dominant risk. These rankings are the thesis-level contribution of the physical-realism extensions, even when the full GPU results are deferred.

The failure-mode atlas can be updated systematically as each approximation is relaxed. Hardware-instance overfitting (the 10.00 % collapse) is *mitigated* by correlated D2D: the degradation is bounded rather than catastrophic. The dual-attention nonlinear-write collapse is *unchanged* by the extensions in this chapter; it is a surrogate-fidelity problem, not a statistics problem. The severe-NL recovery band ($\sim 80\text{--}82\%$ fresh-instance mean across all M-series routes) could be *worsened* by heavy-tailed mismatch or IR drop, because both extensions increase the variance of the effective weights beyond the Gaussian baseline that produced the current band. The retention plateau (79.13–79.51 %) is *downward-shifted* by temperature acceleration and long-horizon extrapolation. This mapping provides a structured way to update the atlas when new device data arrive. It is the final thesis-differentiated contribution of the chapter: not merely to list deferred physics, but to show how each deferred physics layer would reshape the boundaries of the failure modes that the manuscript identified.

6.5 Summary

This chapter has reopened the physical-realism extensions that the manuscript candidly deferred. The contributions are layered:

1. **Correlated D2D** (Section 6.2): the full 10×5 AR(1) sweep is presented and interpreted. Degradation is bounded and monotonic from the i.i.d. baseline to $\rho = 0.5$; see Chapter 7, Table 7.1 for the locked numbers.
2. **Heavy-tailed conductance, IR-drop modeling, temperature drift, and long-horizon retention** (Section 6.3): four “what-if” extensions are defined as extended contributions. Each has a ready code path or protocol (stub script, fixed-point pre-pass, Arrhenius framework, accelerated-aging design), but none has been executed on GPU. They are framed as decision aids that rank which physics layers most urgently need measured device data.

The synthesis (Section 6.4) maps each extension to a contribution tier—core, extended, or future—and connects the extensions back to the failure-mode atlas of Chapter 4. The overarching principle is that the first-order behavioral model is a decision aid, not a chip-predictive emulator. Its value lies in ranking which physics extensions most urgently need measured device data, and in providing the profile-driven interface through which those data can enter the training loop once they arrive.

Surviving physical realism is necessary but not sufficient; the thesis now translates survival into deployment rules.

Chapter 7

Deployment Envelope

7.1 Introduction: from simulation data to shipping criteria

The preceding chapters have established a physical picture of what happens when a modern Vision Transformer is mapped to an organic optoelectronic compute-in-memory (CIM) array. We have defined hardware-aware training (HAT) cadences (Chapter 3), catalogued failure modes (Chapter 4), and quantified recovery under distributional training objectives. The numbers are internally consistent: Ensemble HAT preserves $86.16 \pm 0.19\%$ fresh-instance accuracy under canonical uniform noise; a lower-noise OPECT proxy reaches $88.53 \pm 0.08\%$; severe write nonlinearity ($NL = 2.0$) with the revised gradient-scaling recipe recovers to the $\sim 80\text{--}82\%$ band across Standard, Ensemble, and proportional-noise training routes, leaving a residual gap of roughly 5.7 pp to the canonical $NL = 1.0$ baseline. These are simulation results, and they are bounded by first-order behavioral approximations.

This chapter asks the pragmatic question that every simulation study must eventually face: *so what?* Under which combinations of physical parameters does the Ensemble HAT protocol actually preserve a usable accuracy ranking when the model is dropped onto an unseen hardware instance? Where does that ranking collapse? And if one were advising an analog-CIM program manager—someone deciding whether to tape out a ViT accelerator on an organic RRAM process—what actionable guidance could be distilled from the data?

The chapter is organized as a decision-support document rather than as a conventional results narrative. Section 7.2 constructs a *ranking-preservation map*: a qualitative decision matrix that classifies combinations of D2D variability, nonlinearity severity, ADC precision, retention age, and spatial correlation according to whether the canonical accuracy ordering (Ensemble HAT > fixed-mask HAT > naive deployment) survives on fresh hardware instances. Section 7.3 identifies the complementary *collapse zone*: parameter combinations that destroy the ranking or drive accuracy toward the

chance level. Section 7.4 translates these two maps into a prose decision diagram organized by device-maturity tier (research-grade $\rightarrow$ pilot $\rightarrow$ production). Section 7.5 addresses the architecture question: at low maturity, would a simpler convolutional neural network (CNN) outperform the ViT under the same HAT protocol? Section 7.7 distills industrial-relevance bullets for program managers. Section 7.8 closes with the explicit boundaries of the envelope—what is *not* in the model and therefore not in the recommendation.

Throughout, the treatment is deliberately more applied and systems-oriented than the preceding chapters. Equations are used sparingly; tables and decision matrices carry the argument. Where interpolation or extrapolation is required to bridge gaps in the experimental grid, the reasoning is stated explicitly and the uncertainty is quantified or bounded.

7.2 Ranking-preservation map

The central deployment question for this thesis is not “what is the best possible accuracy?” but “does the training protocol that works in simulation still work when the physical instance changes?” In the language of Chapter 3, the ranking of interest is

$$\text{Ensemble HAT} \succ \text{fixed-mask HAT} \succ \text{naive (no-HAT) deployment}, \quad (7.1)$$

where the comparison is evaluated under the fresh-instance protocol: 10 independent hardware realizations $\times$ 5 Monte Carlo forward passes per realization. A parameter combination *preserves the ranking* if Ensemble HAT maintains a decisive accuracy advantage over fixed-mask HAT, which in turn outperforms naive deployment, on unseen arrays. The ranking is *preserved with degradation* if the ordering holds but the margin between Ensemble HAT and the digital baseline shrinks. It is *collapsed* if fixed-mask HAT equals or exceeds Ensemble HAT on fresh instances, or if any protocol drops to near-chance accuracy.

This definition is stricter than source-domain comparison. Under evaluation on the training-time array, fixed-mask HAT can reach $87.95 \pm 0.27\%$ (the canonical V4 result), which is competitive with Ensemble HAT’s held-out exploratory scan (88.41% in the 50-epoch ablation of Chapter 3). But source-domain accuracy is diagnostically irrelevant for deployment because the deployed array is not the training array. The ranking of Eq. 7.1 is therefore always understood as a fresh-instance statement unless noted otherwise.

7.2.1 Parameter axes and the available data grid

The simulator exposes five physical axes that jointly determine deployment accuracy:

1. **D2D variability amplitude** (σ_{D2D}): canonical 10%, OPECT proxy 3%, sweep grid $\{1, 3, 5, 8, 10, 15, 20\}\%$.

Table 7.1: Green-zone parameter combinations with direct fresh-instance evidence.

Condition	Key parameters	Ensemble HAT	Fixed-mask	Naive
Canonical uniform noise	$\sigma_{\text{D2D}} = 10\%$, i.i.d., $NL = 1.0$	86.16 ± 0.19	10.00 ± 0.00	$\ll 10$
Low-noise literature proxy	$\sigma_{\text{D2D}} = 3\%$, $\sigma_{\text{C2C}} = 2\%$ (OPECT)	88.53 ± 0.08	≈ 10 (cross-profile)	$\ll 10$
Moderate spatial correlation	$\rho = 0.3$, $NL = 1.0$	84.57 ± 2.39	10.00 (extrap.)	$\ll 10$
Strong spatial correlation	$\rho = 0.5$, $NL = 1.0$	82.12 ± 3.95	10.00 (extrap.)	$\ll 10$

2. **D2D spatial structure:** i.i.d. Gaussian versus separable AR(1) correlation with $\rho \in \{0.3, 0.5\}$.
3. **ADC resolution:** sweep grid $\{2, 3, 4, 5, 6, 7, 8, 10, 12\}$ bits.
4. **Write nonlinearity severity** (NL): ideal 1.0, severe 2.0.
5. **Retention age** (t): 0 s (freshly programmed) to 10,000 s under double-exponential drift.

Not every combination in this $7 \times 3 \times 9 \times 2 \times 5$ grid has been evaluated. The canonical Ensemble HAT checkpoint was trained under ($\sigma_{\text{D2D}} = 10\%$, i.i.d., 4-bit programming, $NL = 1.0$, $t = 0$) and then transferred across subsets of the remaining axes. The ranking-preservation map is therefore a *qualitative mosaic* in which some tiles are anchored by direct measurement and others are filled by interpolation or bounded extrapolation.

7.2.2 Green zone: high-confidence ranking preservation

Table 7.1 lists the parameter combinations for which direct fresh-instance data exist and the ranking of Eq. 7.1 is preserved with high confidence.

The OPECT case confirms the ranking persists under a lower-variance literature-calibrated profile ($\sigma_{\text{D2D}} = 3\%$, $\sigma_{\text{C2C}} = 2\%$), with Ensemble HAT reaching $88.53 \pm 0.08\%$ versus fixed-mask collapse to 10.00%. This sets an accuracy ceiling of roughly 88.5% under near-ideal analog conditions—still ~ 9.5 pp below the digital FP32 baseline (98.06%) but ~ 14 pp above standard-HAT collapse. Spatially correlated D2D produces bounded degradation: at $\rho = 0.3$ the Ensemble HAT mean is $84.57 \pm 2.39\%$ (-1.76 pp versus i.i.d.), and at $\rho = 0.5$ it is $82.12 \pm 3.95\%$ (-4.20 pp), with the worst fresh instance at 73.7%—far from chance. The fixed-mask entries for correlated D2D are extrapolated (only Ensemble HAT was evaluated), but there is no physical mechanism by which spatial correlation would restore fixed-mask transfer, so 10.00% is a conservative lower bound.

Table 7.2: Qualitative ranking-preservation matrix. Colors: **G**=green (direct fresh-instance evidence), **Y**=yellow (reasonable extrapolation), **O**=orange (marginal viability), **R**=red (collapse). The matrix is conditional on Tiny-ViT, CIFAR-10, and the canonical 100-epoch Ensemble HAT recipe.

Condition	$NL = 1.0$, ideal write			$NL = 2.0$, severe write	
	ADC ≥ 6	ADC=5	ADC<5	Fresh inst.	Source only
Low D2D ($\leq 3\%$)	G	Y	R	O (extrap.)	O
Canonical D2D (10 %)	G	O	R	R (extrap.)	O
High D2D (15–20 %)	Y	O	R	R (extrap.)	R (extrap.)
$\rho = 0.3$ correlated	G	Y	R	R (extrap.)	O (extrap.)
$\rho = 0.5$ correlated	G	Y	R	R (extrap.)	O (extrap.)
Retention $t > 10$ s + recal.	Y	Y	R	Y (extrap.)	Y

Table 7.2 condenses the four zones into a single qualitative matrix. **Green:** direct fresh-instance evidence (canonical 10 % D2D i.i.d., OPECT 3 % D2D, correlated D2D up to $\rho = 0.5$). **Yellow:** plausible extrapolations where the ranking likely holds but margins narrow or data are missing—retention drift with dynamic scale recalibration (accuracy floor near 79 %, uncertainty ± 3 –5 pp), proportional-noise HAT (plausible low-90 % fresh-instance mean, uncertainty ± 5 –8 pp), and inverse-gamma front-end compensation (+5.8 pp at $\gamma_{\text{phys}} = 2.0$). **Orange:** marginal viability—severe write nonlinearity ($NL = 2.0$) produces a consistent ~ 80 –82 % band across all M-series routes (tight variance < 2 pp), and the ADC 5-to-6-bit boundary shows a ~ 7 pp jump with a critical threshold near 5.5 bits (± 0.5 bits uncertainty). **Red:** collapsed ranking—ADC < 5 bits (near-chance regardless of training), fixed-mask HAT on any fresh instance (10.00 %), and high-D2D ($> 15\%$) combined with severe NL or sub-5-bit ADC.

The key asymmetry is that the ranking is robust to moderate D2D increases and spatial correlation, but fragile to ADC precision reductions and severe write nonlinearity. A program manager should therefore prioritize peripheral readout precision and programming linearity over incremental device-to-device uniformity improvements.

7.3 Collapse zone

The complement of the ranking-preservation map is the *collapse zone*: the set of parameter combinations that destroy the accuracy ordering of Eq. 7.1 or drive all protocols to near-chance performance. Identifying the collapse zone is as important as identifying the safe zone, because it tells a program manager which physical specifications are *non-negotiable* and which are merely performance-negotiable.

Three collapse boundaries are unambiguous in the data. First, the ADC precision cliff: below 5-bit ADC, accuracy is near-chance ($\sim 10\%$) regardless of D2D level or

Table 7.3: Collapse zone: qualitative danger-zone specification. Entries marked “extrap.” are not directly measured but are inferred from the monotonic trends of the individual sweeps.

Trigger combination	Expected outcome	Confidence
ADC < 5 bits, any D2D/NL	Near-chance ($\sim 10\%$)	High (direct)
Fixed-mask HAT, any fresh instance	10.00 % collapse	High (direct)
$NL \geq 2.0$, fresh instance	$\leq 32\%$ ceiling	High (direct)
$NL \geq 2.0 + \rho \geq 0.5 + \text{retention} > 1 \text{ week}$	Likely < 25 %	Medium (extrap.)
$NL \geq 2.5$, any fresh instance	Likely < 20 %	Medium (extrap.)
ADC = 5 bits + high D2D (> 15 %)	30–50 % (non-viable)	Medium (interpol.)
No dynamic recalibration + retention > 10 s	Accuracy drifts unbounded	Medium (extrap.)

training protocol, because quantization noise swamps the signal before any mismatch is applied (Eq. 3.1). The 5-to-6-bit transition yields a consistent ~ 7 pp jump; above 6 bits degradation becomes D2D-dominated. Second, fixed-mask HAT collapses to $10.00 \pm 0.00\%$ on every fresh instance under both canonical and OPECT noise profiles—the memorization of a specific M_0 is independent of $p(M)$, so lower D2D or correlation does not help. Fixed-mask HAT is therefore not a deployment candidate for any process requiring fresh-instance robustness. Third, severe write nonlinearity ($NL = 2.0$) drives all training routes (Standard, Ensemble, proportional-noise) into a tight $\sim 80\text{--}82\%$ band with cross-instance variance < 2 pp, indicating NL has become the binding constraint and the marginal benefit of epoch resampling is masked. The residual gap to the $NL = 1.0$ baseline (~ 5.7 pp) is systematic, not catastrophic.

No experiment evaluates the joint application of severe NL, correlated D2D, and retention drift. Table 7.3 assembles the individual collapse boundaries into a qualitative danger-zone specification.

The combination $NL > 2.5 + \rho > 0.5 + \text{retention} > 1 \text{ week}$ is the archetypal danger zone: none of the three conditions alone collapses Ensemble HAT to chance, but their joint effect is expected to be super-additive, with a conservative extrapolation below 25 % (medium confidence).

7.4 Decision diagram: should this architecture ship?

The ranking-preservation map and the collapse zone can be combined into a prose decision diagram. The branching criterion is *device maturity tier*, a qualitative label

that maps onto the physical parameter ranges tested in the simulator.

Research-grade arrays are characterized by $\sigma_{\text{D2D}} \geq 10\%$ (often 15–20%), write nonlinearity $NL \geq 1.5$, ADC resolution ≤ 5 bits or uncalibrated, and retention time constants in the hundreds of milliseconds without on-chip recalibration circuitry. At this maturity level, the collapse-zone boundary is crossed for at least one non-negotiable parameter: sub-5-bit ADC produces near-chance accuracy regardless of training protocol, severe NL drives fresh-instance accuracy below 32% even with MLP linearization, and the retention plateau near 79% is only reachable if dynamic scale recalibration is available, which typically requires reference-cell readout circuitry that research-grade prototypes lack. The recommended use of such arrays is as a *training-diagnostic platform* rather than a deployment target: train Ensemble HAT checkpoints, evaluate them on the research-grade array, and use the gap between source-domain and fresh-instance accuracy to guide process development. If a simpler task or smaller model is required, consider a shallow CNN (Section 7.5), although fresh-instance CNN data under Ensemble HAT are not available in the present dataset. The single highest-leverage intervention at this tier is improving the ADC from 5 to 6 bits, which moves the system from the red zone to the yellow zone of Table 7.2.

Pilot-line arrays are characterized by $\sigma_{\text{D2D}} \in \{3\%, 5\%, 8\%\}$, $NL \in \{1.0, 1.2\}$ (ideal to mildly nonlinear), 6–8-bit calibrated ADC, and basic on-chip reference cells that permit intermittent scale recalibration. At this tier, the canonical ranking is preserved with commercially relevant margins: the OPECT proxy (3% D2D, 2% C2C) reaches $88.53 \pm 0.08\%$, only ~ 9.5 pp below the digital baseline, while the canonical 10% D2D regime reaches $86.16 \pm 0.19\%$, viable for many edge-classification tasks. The 6-bit ADC cliff has been cleared, and the retention plateau near 79% is accessible provided recalibration is performed at intervals shorter than ~ 1 s when the application demands $> 82\%$ accuracy. The deployment checklist is straightforward: train with per-epoch D2D resampling (Ensemble HAT), not fixed-mask HAT; characterize the physical photoresponse exponent γ_{phys} and deploy inverse-gamma compensation if $\gamma_{\text{phys}} > 1$ (expected gain +5.8 pp at $\gamma_{\text{phys}} = 2.0$); include dynamic scale recalibration in the inference stack and budget peripheral energy for reference-cell readout; validate fresh-instance robustness on at least 10 physical arrays before mass programming; and if spatial correlation is suspected from wafer maps, treat the $\rho = 0.3$ result (84.57%) as a conservative accuracy floor and the $\rho = 0.5$ result (82.12%) as a stress-test bound.

Production-grade arrays are characterized by $\sigma_{\text{D2D}} \leq 3\%$, $NL \approx 1.0$ (near-ideal write or iterative write-verify), ≥ 8 -bit ADC, and stable retention with on-chip recalibration. At this tier, the analog noise floor is low enough that even fixed-mask HAT might achieve acceptable source-domain accuracy, but the OPECT case study demonstrates that fixed-mask checkpoints still collapse when transferred across profiles. Hardware-instance overfitting therefore does not vanish automatically at low noise, and Ensemble HAT should remain the default training protocol, though its role shifts from “rescue” to “insurance.” The expected fresh-instance accuracy is in the high-80% to low-90% range, with a tight variance (± 1 pp or better) that simplifies yield

planning: at $88.53 \pm 0.08\%$, the cross-instance standard deviation is smaller than typical test-set sampling noise, so yield can be planned on the *mean* rather than a worst-case tail. If the application threshold is 85% , virtually every instance passes; if the threshold is 87% , the yield is still $> 99\%$ under a Gaussian tail assumption. The energy estimate ($\approx 11\times$ dense-projection reduction versus FP32) becomes credible at this tier because the peripheral overhead is dominated by calibrated digital circuits rather than speculative analog compensation.

For quick reference, the decision logic can be summarized as a compact branching tree:

1. **ADC ≥ 6 bits?**

- No $\rightarrow$ DO NOT SHIP. Invest in peripheral readout.
- Yes $\rightarrow$ continue.

2. **Write nonlinearity $NL < 1.5$?**

- No $\rightarrow$ DO NOT SHIP ViT. Consider simpler model or process repair.
- Yes $\rightarrow$ continue.

3. **Device maturity tier?**

- Research-grade (high D2D, no recalibration) $\rightarrow$ DIAGNOSTIC USE ONLY.
- Pilot (moderate D2D, recalibration available) $\rightarrow$ SHIP WITH ENSEMBLE HAT + COMPENSATION.
- Production (low D2D, ideal write, 8+ bits) $\rightarrow$ SHIP WITH ENSEMBLE HAT AS INSURANCE.

7.5 CNN versus ViT at different maturity tiers

The deployment envelope developed above is anchored to Tiny-ViT-5M, a compact Vision Transformer fine-tuned from ImageNet. A natural question arises: at low device maturity, where noise and nonlinearity are most severe, would a simpler convolutional architecture outperform the transformer under the same HAT protocol? This section addresses the question with the evidence available and flags the gaps.

The manuscript reports three architecture–accuracy points under matched digital training: ResNet-18 (95.46%), ConvNeXt-Tiny (90.74%), and Tiny-ViT-5M (98.06%) on CIFAR-10. Under the canonical analog noise model with standard HAT, the ranking is preserved—Tiny-ViT V4 reaches $87.95 \pm 0.27\%$, while ConvNeXt C4 reaches a comparable level—but *no fresh-instance Ensemble HAT sweep has been performed for ConvNeXt or ResNet*. The CNN numbers are either source-domain or single-run estimates, and the 10×5 fresh-instance protocol has not been applied to a CNN testbed. The manuscript Discussion notes that “the Transformer architecture is more

fragile under the present physical stress tests,” because in a CNN local convolution and pooling average perturbations spatially and ReLU saturation limits the propagation of intensity-dependent scaling errors, whereas in a ViT patch embeddings are fed directly into global self-attention where a sublinear photoresponse alters the token embedding and the softmax exponentiates similarity changes across the entire sequence without spatial averaging. This structural difference predicts that the ViT should degrade more sharply than a CNN under front-end distortion and spatially correlated D2D.

At the research-grade tier—high D2D, high NL, sub-5-bit ADC—neither architecture is deployable; the ADC cliff is architecture-agnostic. The question becomes interesting only at the boundary of viability. Consider the pilot tier with moderate D2D (5–10 %) and a 6-bit ADC: the CNN’s spatial averaging provides built-in regularization against structured mismatch, because a 3×3 convolution with stride 2 averages nine spatial locations before the nonlinearity, effectively low-pass filtering the D2D field. The ViT patch embedding is also a convolution, but the subsequent attention layers operate on token vectors that have *already been corrupted* by the full spatial mismatch pattern, and there is no spatial averaging in the attention mechanism. One therefore expects the CNN to tolerate higher D2D or stronger spatial correlation at a given accuracy level. Similarly, under severe write nonlinearity ($NL = 2.0$), the dual-attention collapse documented in Chapter 4 is a *transformer-specific* failure mode; a pure CNN has no QKV or output-projection analog paths to linearize, so if the severe-NL bottleneck is concentrated in the MLP path, a CNN might reach the MLP-only source-domain rescue ($\approx 88\%$) without the attention-side collapse, and its fresh-instance ceiling might exceed the ViT’s 32 % bound.

These arguments are theoretical, not empirical. The absence of a fresh-instance CNN comparison is the most important data gap in the deployment envelope. A direct experiment would train ConvNeXt (or ResNet-18) with Ensemble HAT under the canonical uniform-noise profile and evaluate on 10 fresh instances. If the CNN fresh-instance mean exceeds 86.16 %, the ViT is not the optimal architecture for organic CIM at this maturity level; if the CNN mean is comparable or lower, the ViT’s superior digital baseline justifies its higher fragility. Until such data exist, the conservative recommendation is to prefer the ViT at production maturity (low noise, ideal write) for its proven Ensemble HAT transfer (88.53 %); to deploy the ViT at pilot maturity but consider a CNN as a process-margin hedge if D2D or correlation is worse than expected; and to recognise that at research maturity architecture choice is moot because the process is not ready for either. This framing aligns with the broader CIM literature: recent heterogeneous analog accelerators for Vision Transformers, including HARDSEA⁶, adopt the same hybrid analog–digital partition but target more mature CMOS-under-ReRAM processes, suggesting that CNN-first deployment may be a pragmatic stepping stone toward eventual ViT deployment on organic arrays.

Table 7.4: Cross-architecture proportional-noise HAT validation on TinyImageNet-200 (fresh-instance mean, $n = 10$ realisations $\times$ 5 MC runs per realisation).

Architecture	HAT	Seed	Source (%)	Fresh (%)	Std (%)
DeiT-Small	digital	123	48.22	48.22	0.00
DeiT-Small	proportional	123	50.24	50.20	0.10
DeiT-Small	digital	456	53.61	53.61	0.00
DeiT-Small	proportional	456	54.44	54.19	0.09
DeiT-Small	digital	789	53.58	53.58	0.00
DeiT-Small	proportional	789	56.54	56.33	0.13
ViT-Small	digital	123	48.83	48.83	0.00
ViT-Small	proportional	123	49.03	49.00	0.09
ViT-Small	digital	456	54.58	54.58	0.00
ViT-Small	proportional	456	54.06	53.90	0.13
ViT-Small	digital	789	50.86	50.86	0.00
ViT-Small	proportional	789	55.85	55.41	0.10

7.6 Cross-architecture validation

The deployment envelope above is anchored to Tiny-ViT-5M on CIFAR-10. A critical unanswered question is whether the ranking logic—Ensemble HAT $\succ$ fixed-mask HAT $\succ$ naive deployment—survives the transition to a larger dataset and to different transformer architectures. To test this, a cross-architecture validation was performed on TinyImageNet-200 (200 classes, 64×64 resolution) using DeiT-Small and ViT-Small (both Patch16-224 backbones). The protocol matches the thesis fresh-instance standard: 10 independent hardware realizations $\times$ 5 Monte Carlo forward passes per realization.

The cross-architecture results are summarised below. Table 7.4 reports the per-seed fresh-instance accuracy for digital and proportional-noise training on both architectures. Three patterns are immediately apparent.

First, proportional-noise HAT eliminates fresh-instance mode collapse. On every seed and both architectures, proportional training reaches fresh-instance accuracy comparable to the digital baseline (fresh-source gap -0.04 to -0.44 pp), with no evidence of the catastrophic collapse that afflicts Standard HAT (≈ -34 pp relative to train accuracy). Across three seeds, proportional HAT outperforms digital training by a mean of $+1.77$ pp on DeiT and $+1.35$ pp on ViT (fresh-instance mean). This confirms that the proportional noise law—which scales mismatch variance with conductance magnitude—is not merely a physically plausible stress test but a robust drop-in alternative that can *improve* accuracy when the device physics support it.

Second, the severity of Standard-HAT collapse is architecture-general. In separate early-seed experiments (not shown in Table 7.4), DeiT drops from 40.9% train to 6.6% fresh (-34.3 pt) and ViT drops from 38.8% to 6.9% (-31.9 pt) under fixed-mask Standard HAT, indicating that hardware-instance overfitting is a property

of the *training protocol*, not of a particular backbone.

Third, Ensemble HAT partially mitigates but does not fully eliminate the collapse. On TinyImageNet-200, Ensemble HAT reaches 40.78 % (DeiT) and 40.16 % (ViT), roughly -4 pt below train accuracy, a meaningful improvement over Standard HAT but less complete than the 86.16 % recovery on CIFAR-10. The gap reflects dataset scale, resolution, and architecture capacity, not a failure of the resampling principle itself.

The cross-architecture data do not change the green-zone ranking, but they *broaden* its scope. The canonical accuracy ordering was derived on a single architecture and dataset; the TinyImageNet-200 validation shows that the *direction* of the ordering survives the transition to larger vision tasks, even if the *absolute margin* shrinks. Proportional HAT, which was treated as a physical-extension stress test in the CIFAR-10 experiments, emerges as a credible primary training protocol for regimes where the noise law is known to be conductance-dependent.

An important caveat is seed sensitivity. The digital and proportional protocols exhibit non-trivial seed variance (± 2.5 – 2.9 pt across seeds on DeiT, ± 0.2 – 4.6 pt on ViT), with the ViT digital seed456 result (54.58 %) standing $+5.75$ pt above seed123 and $+3.72$ pt above seed789. The outlier is attributable to seed-dependent training dynamics, not a protocol deviation. By contrast, Ensemble and Standard HAT show tight variance (< 0.6 pt) once the analog noise path is active. This pattern suggests that analog noise itself acts as a variance regulariser—once HAT is active, seed-to-seed differences are suppressed by the forward-path perturbation—so the analog-mapped model is no *more* sensitive to training stochasticity than its digital counterpart, and in some regimes it is less so.

7.7 Implications for analog-CIM program managers

The deployment envelope is not merely an academic classification; it is a tool for resource allocation. This section translates the findings into industrial-relevance bullets for program managers deciding how to invest non-recurring engineering (NRE) dollars across device fabrication, peripheral design, and training infrastructure.

For program managers, the deployment envelope translates into four concrete take-aways. First, yield is D2D-limited, not C2C-limited: once the ADC is ≥ 6 bits, D2D variability accounts for 92.2 % of residual accuracy variance. The i.i.d. and $\rho = 0.3$ regimes are safe for applications requiring > 80 % accuracy, but $\rho = 0.5$ introduces a tail risk in which one instance in ten could drop below 75 %; wafer-scale spatial covariance should therefore be measured and wafers with $\rho > 0.4$ rejected unless architectural mitigations are implemented. Second, Ensemble HAT eliminates per-device retraining, saving roughly 20,000 GPU-hours (or \$5–10k in cloud compute at 2025 rates) for a pilot line of 100 test arrays. Third, the retention plateau at 79.13–

79.51 % dictates a task-dependent refresh specification: high-accuracy applications need refresh intervals < 1 s, moderate-accuracy sensing can tolerate 10–100 s, and ultra-low-power intermittent inference can recalibrate only at wake-up. The peripheral energy for reference-cell readout is not included in the manuscript’s 11.45 $\times$ energy-reduction estimate; a realistic model would subtract 5–15 % as a function of refresh frequency. Fourth, the 11.45 $\times$ figure itself is an upper bound realisable only at production maturity, where peripheral circuits are dominated by calibrated digital logic. Finally, the compound stress test (89.61 % under simultaneous 2 % C2C, 3 % D2D, 6-bit ADC, 1 % IR drop, 1 % sneak path, and 1,000 s retention) provides a “smoke test”: if a target profile predicts below 80 % under comparable joint stress, the design is too aggressive for tape-out.

7.8 Limitations of the envelope

Every deployment recommendation in this chapter is conditional on the first-order behavioral approximations of Chapter 2. The envelope is a *decision aid*, not a predictive emulator. The following phenomena are not in the model and therefore not in the recommendation, with their expected impacts on safe-zone boundaries:

- **Spatial IR drop:** The framework uses a 1 % scalar placeholder, but real IR drop is geometry-dependent and produces deterministic position-dependent weight distortion. Large arrays (512×512) may see effective $\sigma_{\text{eff}} > 15$ % at the corners; SPICE-calibrated parasitic models (e.g., CrossSim²) should bracket this before relying on the 10 % D2D safe zone.
- **Sneak paths:** The 1 % scalar placeholder may understate pattern-dependent, non-Gaussian crosstalk. Multiplicative errors from unselected cells break the scale-recovery assumption and could shift the 6-bit ADC cliff upward. Safe-zone boundaries are unknown until selector or 1T1R devices are characterized.
- **Inter-tile coupling and temperature:** The simulator treats each layer as an independent tile, omitting routing resistance, capacitive loading, and thermal gradients. The OPECT 88.53 % prediction is optimistic if array geometry adds position-dependent spread. Temperature is a major omission: organic semi-conductors are strongly temperature-sensitive, and the 79 % retention plateau could shift downward significantly above 25°C. A full temperature sweep is the highest-priority extension.
- **Joint failure-mode interactions:** All failure modes were evaluated in isolation. The compound stress test (89.61 %) is a single point in the joint space; whether retention drift increases hardware-instance overfitting susceptibility, whether the 32 % severe-NL ceiling shifts under correlated D2D, and whether the 6-bit

ADC cliff moves with front-end distortion are all unanswered. These define the extrapolation boundary.

- **Surrogate fidelity:** The $NL = 2.0$ boundary is a property of the $(G_{\max} - G)^{NL-1}$ gradient-scaling surrogate, not necessarily of physical ionic-migration dynamics. The envelope should be recalibrated when measured write data become available; the profile-driven interface (Chapter 2) supports this without code changes.

7.9 Summary

This chapter has converted the simulation data of the preceding chapters into an actionable deployment envelope for organic optoelectronic CIM. The central construct is the *ranking-preservation map* (Section 7.2), which classifies parameter combinations according to whether the canonical accuracy ordering—Ensemble HAT > fixed-mask HAT > naive deployment—survives on unseen hardware instances. Direct fresh-instance evidence exists for the canonical uniform-noise regime ($86.16 \pm 0.19\%$), the low-noise OPECT proxy ($88.53 \pm 0.08\%$), and spatially correlated D2D up to $\rho = 0.5$ ($82.12 \pm 3.95\%$). Extrapolation with explicit uncertainty bounds extends the map to retention drift ($\approx 79\%$ plateau), proportional-noise regimes (plausible low-90 % fresh-instance mean), and the ADC 5-to-6-bit boundary.

The *collapse zone* (Section 7.3) identifies non-negotiable boundaries: $ADC < 5$ bits produces near-chance accuracy regardless of training; fixed-mask HAT collapses to 10.00 % on every fresh instance; severe write nonlinearity ($NL = 2.0$) creates a 32 % fresh-instance ceiling even with MLP linearization. The joint danger zone $NL > 2.5 + \rho > 0.5 + \text{retention} > 1 \text{ week}$ is extrapolated to fall below 25 %, with medium confidence.

The *decision diagram* (Section 7.4) branches on device maturity tier. Research-grade arrays should be restricted to diagnostic use; pilot-line arrays can ship with Ensemble HAT plus front-end compensation; production-grade arrays can treat Ensemble HAT as insurance rather than rescue. The CNN-versus-ViT comparison (Section 7.5) remains an open question for fresh-instance CNN data, but theoretical reasoning suggests that CNNs may tolerate higher D2D and NL at the pilot-tier boundary.

For *program managers* (Section 7.7), the key takeaways are: prioritize spatial uniformity over C2C repeatability; use Ensemble HAT to save per-device retraining NRE; negotiate the retention refresh interval with the application team before finalizing peripheral energy budgets; and treat the compound stress test as a smoke-test benchmark.

The *limitations* (Section 7.8) are substantial and explicitly enumerated: spatial IR drop, sneak-path currents, array geometry, full temperature sweep, and the unexplored interaction frontier. The envelope is valid only within the first-order behavioral approximations of the present framework, and it should be recalibrated when measured

device data become available. Within those bounds, however, it provides a principled basis for deciding whether a given organic RRAM process is ready to host a modern vision backbone—and what training protocol it will need if it is.

The envelope is where the thesis ends and the next project begins.

Chapter 8

Outlook and Conclusion

8.1 Introduction: the thesis as a bridge

This thesis began with a simple observation. Organic optoelectronic compute-in-memory devices are invariably introduced through figures of merit—conductance window, retention time, write asymmetry, optical responsivity—that are necessary but not sufficient. They do not answer the question that ultimately matters for deployment: *what accuracy survives once a modern vision backbone is mapped to an analog array with structured mismatch, finite readout precision, and a sub-linear photoresponse?* The intervening chapters have built a bridge between the measurement laboratory and the algorithm training loop. Chapter 2 constructed the bridge’s truss: a profile-driven simulation framework that turns device characterizations into structured training inputs. Chapter 3 mapped the traffic rules: a taxonomy of hardware-aware training cadences, noise profiles, and transfer guarantees. Chapter 4 surveyed the load limits: a catalogue of named, reproducible failure modes that bound the deployment envelope.

The preceding chapter constructed the deployment envelope; this chapter steps outside it to propose the next-generation theory and hardware roadmap.

This final chapter has two tasks. The first is to look forward. It identifies the experiments, architectures, hardware roadmaps, and theoretical tools that can extend the bridge beyond the scope of the present thesis. The second is to look backward, integrating the eight chapters into a single narrative arc with three core contributions: (1) `computeViT` as a reusable research instrument; (2) the HAT taxonomy and the discovery of hardware-instance overfitting; and (3) a quantified deployment envelope for organic CIM. The outlook is optimistic but grounded; the conclusion is definitive but open-ended—because a thesis, like the bridge it builds, is meant to be crossed.

Table 8.1: Provisional Fresh-D2D audit of selective analog KV-cache. Candidate 107 rows for a 125 M-parameter language model with 5 independent D2D seeds per evaluation level. Train D2D = 0.02 for all selective checkpoints. These values are pilot/audit-only until the 107 manifest or minimal rerun gate supplies per-row provenance and checkpoint hashes.

Checkpoint	Analog layers	Eval D2D	Mean PPL	Std
Digital baseline	—	—	15.68	—
last1 selective (train D2D= 0.02)	[23]	0.02	18.42	0.02
last1 selective (train D2D= 0.02)	[23]	0.04	18.55	0.03
last1 selective (train D2D= 0.02)	[23]	0.05	18.60	0.03
last2 selective (train D2D= 0.02)	[22, 23]	0.02	18.71	0.02
last2 selective (train D2D= 0.02)	[22, 23]	0.04	19.07	0.04
last2 selective (train D2D= 0.02)	[22, 23]	0.05	19.21	0.04
all-layer (train D2D= 0.02)	all	0.05	66.84	5.91

8.2 Immediate next steps: thesis-completion experiments

Two experiments complete the loops opened in Chapter 4. First, **joint severe-NL and instance-overfitting mitigation**: starting from the canonical Ensemble HAT checkpoint ($86.16 \pm 0.19\%$), apply a weights-only warm-start with MLP blocks retaining $NL = 1.0$ linear surrogates while attention blocks see full $NL = 2.0$ distortion, with D2D resampled every epoch. The target is $\geq 80\%$ fresh-instance accuracy under the 10×5 protocol; anything below 50% would signal a fundamental incompatibility between first-order NL surrogates and distributional training. Second, **ImageNet-100 pilot**: test whether the ranking logic (Ensemble HAT $>$ fixed-mask HAT; bounded degradation under correlated D2D) survives dataset scale under the G-AA2 protocol. Success is qualitative—ranking preservation, not headline numbers.

8.3 Language-model CIM: transferring HAT principles beyond vision

The dominant transformer workload is the large language model. Analog CIM offers the same theoretical energy advantage for LLMs, but the mapping is complicated by tiling scale, variable-length attention, and KV-cache storage. A provisional pilot on a 125 M-parameter model (Table 8.1) tested whether *selective* terminal-layer analog KV storage could pass a cross-instance robustness gate. The current 107 evidence is audit-only: it lacks a signed per-row metadata manifest or minimal rerun with corrected-noise provenance, so the table is used here as an outlook target rather than as a locked Paper2 claim.

The provisional pilot suggests a design rule to test, not a locked Paper2 conclusion: **begin with terminal-layer selective mapping, not all-layer mapping.** At eval $D2D = 0.05$, the all-layer candidate degrades to 66.84 ± 5.91 PPL ($4.3\times$ the digital baseline), whereas last1 selective is 18.60 ± 0.03 PPL with cross-instance standard deviation below 0.03 PPL. Last1 also outperforms last2 at every tested D2D level in this audit table, suggesting that earlier layers may amplify perturbations through multiple downstream paths. The core thesis principle—train over the deployment distribution, not a single instance—therefore motivates the next 107 rerun/manifest path; the practical challenge is tile-granularity resampling cost, which motivates mismatch resampling only for analog-mapped layers and low-rank surrogate approximations.

8.4 Hardware-in-the-loop roadmap: from simulation to silicon

The ultimate goal is to deploy HAT-trained models on physical organic arrays. A three-stage roadmap de-risks the transition. **Stage I** replaces first-order surrogates with measured device profiles (spatial covariance, pulse-accurate transients, temperature-dependent retention); the blocker is batch-to-batch variability exceeding within-batch D2D spread. **Stage II** emulates the mixed-signal stack on FPGA to validate control logic; the blocker is resource scaling, since a 256×256 crossbar consumes most of a mid-range FPGA. **Stage III** fabricates a small organic RRAM array (64×64 or 128×128) for functional verification; ecological blockers (humidity, oxygen ingress, contact resistance) dominate. Calibration infrastructure—per-column ADC calibration, reference-cell tracking, temperature compensation—cuts across all stages and is not yet in the energy model.

8.5 Theory directions: from empirical recipes to principled guarantees

Three directions can transform the empirical recipes into principled guarantees. First, **HAT as a regulariser**: epoch-level resampling convolves empirical risk with the mismatch distribution, smoothing sharp minima. The expected gradient variance bounds the Hessian trace at convergence:

$$\theta_{\text{ens}}^* = \arg \min_{\theta} \mathbb{E}_{M \sim p(M)} \left[\frac{1}{N} \sum_{n=1}^N \ell(f(x_n; \theta, M), y_n) \right], \quad (8.1)$$

$$\text{Tr } H(\theta_{\text{ens}}^*) \leq \frac{C}{\sigma_M^2} \mathbb{E}[\|\nabla_{\theta} \ell\|^2], \quad (8.2)$$

where larger σ_M^2 forces a flatter minimum, explaining why Ensemble HAT preserves $86.16 \pm 0.19\%$ fresh-instance accuracy while fixed-mask HAT collapses to 10.00% .

Second, **ensemble-frequency effective width**: define $W_{\text{eff}} = E/\tau_{\text{mix}}$, with E training epochs and τ_{mix} optimizer mixing time. Per-epoch resampling gives $W_{\text{eff}} \approx E$; per-batch gives comparable width but shorter mixing, producing the non-stationary gradient field that empirically underperforms. The inverted-U prediction matches the Chapter 3 ablation.

$$W_{\text{eff}} = \frac{E}{\tau_{\text{mix}}}, \quad (8.3)$$

Third, **PAC-Bayes bounds for device-instance generalisation**: treating $p(M)$ as a prior and learned weights as posterior $q(\theta)$,

$$R_{\text{fresh}}(q) \leq \hat{R}_{\text{ens}}(q) + \sqrt{\frac{\text{KL}(q\|p) + \ln(2\sqrt{E}/\delta)}{2(E-1)}}, \quad (8.4)$$

where the bound tightens as $O(1/\sqrt{E})$, justifying longer training, and the KL term penalises complex posteriors, justifying weight decay. Future tightening includes Gaussian-process priors for spatial correlation and extension to tiled LLM settings.

8.6 Conclusion: one story

A thesis is often read as a sequence of chapters, each with its own figures, equations, and contributions. But the true product is not the sequence; it is the *story* that runs through it. This concluding section tells that story in three acts, then gathers its threads into a single claim about what this thesis has built and where it leads.

The story opens with a gap. Device scientists publish conductance windows and retention curves; system engineers need deployment accuracy. Chapter 2 closed that gap by building **compute-ViT**, a profile-driven simulation framework in which every physically relevant assumption enters through a structured configuration object. The framework is not merely an implementation detail; it is a *reusable research instrument*. A literature-derived prior and an in-house measured profile populate the same fields and drive the same training scripts, so switching from one device generation to the next requires only replacing the profile instance.

With the instrument in hand, Chapter 1 asked the obvious question: if we train a vision transformer on a noisy analog array, does it survive deployment? The answer was a stark $10.00 \pm 0.00\%$ collapse. Standard hardware-aware training, which fixes a single device-to-device mismatch field for the entire training run, overfits that instance as if it were a hidden domain label. The discovery is not that analog noise hurts accuracy—that was known—but that *robustness to analog noise is not the same as robustness to hardware-instance shift*. The remedy, Ensemble HAT, is simple in retrospect: resample the mismatch field at epoch boundaries so that the optimizer

learns over a *distribution* of arrays rather than memorising one sample from it. The result— $86.16 \pm 0.19\%$ fresh-instance accuracy—is the empirical anchor of the thesis. Chapter 3 placed that anchor in a broader design space. Ensemble HAT is not an isolated trick; it is one point in a taxonomy of resampling cadences, noise profiles, and transfer guarantees. Per-epoch resampling outperforms per-batch because it creates a block-stationary optimisation landscape: stable enough to descend within each block, diverse enough to generalise across blocks. These distinctions provide a *language* for discussing HAT that did not exist before.

If the first act built the instrument and discovered the central phenomenon, the second act mapped its boundaries. Chapter 4 organised the empirical landscape as a structured catalogue of named, reproducible breakdown patterns. The atlas is not a pessimistic document; it is a *deployment envelope*. It says: if your device operates at $NL = 1.0$ with i.i.d. mismatch, you can expect $86.16 \pm 0.19\%$ on fresh instances. At $\rho = 0.3$, the mean is $84.57 \pm 2.39\%$; at $\rho = 0.5$, it drops modestly to $82.12 \pm 3.95\%$ —bounded degradation, not catastrophic collapse. If retention drift is present but scale recalibration is applied, the accuracy stabilises in the $79.13\text{--}79.51\%$ plateau. These numbers are contracts between physics and algorithm. A cross-cutting theme binds the atlas entries together: the primacy of *distributional robustness* over instance-specific rescue. Standard HAT rescues source-domain accuracy; MLP-linearisation rescues source-domain accuracy under severe NL; but neither survives fresh-instance evaluation. Only interventions that explicitly average over the deployment distribution produce results that are stable across multiple hardware realisations. This principle is the methodological legacy of the thesis.

The third act is the outlook developed in the preceding sections of this chapter. The joint MLP-linear + Ensemble HAT experiment tests whether the two deepest failure modes can be simultaneously repaired. The ImageNet-100 pilot tests whether the ranking logic survives dataset scale. The language-model direction tests whether the principles transfer to the dominant AI workload of the coming decade. The hardware roadmap lays out a staged path from simulation to silicon. And the theory directions offer a route from empirical recipes to principled guarantees. These threads are not disjoint futures; they are convergent. A language-model HAT protocol will need the regularisation theory to choose resampling cadences across thousands of tiles; a chip prototype will need the FPGA stage to validate calibration logic; and every stage will need the profile-driven framework as its common language. The thesis therefore does not end with a single result; it ends with a *platform*.

It is worth stating the contributions plainly, because a thesis that spans simulation, algorithm, and hardware can easily be misread as three small papers bound together. It is not. The contributions are three facets of one story. First, **compute-ViT** is a reusable research instrument: the first simulation environment designed specifically for organic optoelectronic CIM that treats device profiles as interchangeable structured inputs, with pluggable physical interfaces, PyTorch autograd integration, and external validation against CrossSim. Second, the HAT taxonomy formalises the

design space of resampling cadences and noise profiles, and the instance-overfitting discovery redefines the evaluation criterion for analog-deployment experiments—the fresh-instance protocol (10 arrays $\times$ 5 Monte Carlo evaluations) is a minimal bar for claiming deployment-grade robustness. Third, the failure-mode atlas provides the first system-level accuracy contracts for Vision Transformers on organic optoelectronic arrays, telling the materials scientist which device properties matter most, the circuit designer which peripheral overhead is acceptable, and the algorithm designer which training protocol to choose for a given physical regime.

This thesis does not claim that organic optoelectronic compute-in-memory is ready for ImageNet-scale deployment today. The severe-NL residual gap, the retention plateau, and the environmental sensitivity of organic devices are real blockers that will require years of joint materials–circuit–algorithm research to overcome. What the thesis claims is something more modest and, perhaps, more durable: that the transition from “device works” to “system works” requires a new kind of experimental science. That science must be *profile-driven*, treating device characterisations as structured inputs rather than narrative context. It must be *distributional*, training over the deployment distribution rather than a single hardware sample. And it must be *transfer-oriented*, evaluating every intervention on fresh instances before declaring it deployment-grade. The clean room is quiet and the fabrication tools are precise, but the real world is a distribution of arrays, temperatures, and inference latencies that no single measurement can capture. Building a bridge between those two worlds is slow, incremental, and often humbling. Yet every time a trained model survives a fresh hardware instance—every time the accuracy stays above 80 % on an array it has never seen—the bridge lengthens. This thesis has lengthened it a little. The path ahead is to keep building.

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